Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

Prenominal Adverbs, or Apparent Selectional Violations in Coordination

Adam Przepiórkowski
Agnieszka Patejuk

A robust generalization about coordination is that X and Y may be conjoined in position P if and only if X and Y may each occur alone in P. In particular, different categories may be coordinated in a position that allows either. Apparent violations of this generalization are sometimes taken as evidence for an asymmetric structure of coordination, where one conjunct determines categorial features of the coordinate structure. Bruening and Al Khalaf (2020) and Bruening (2023) argue that one such violation involves coordination of AdvPs and AdjPs in prenominal positions that apparently do not allow AdvPs alone, claiming that *The [Once and Future] King* is grammatical, while *the once king* is either ungrammatical or involves a hypothetical compound, *once king*, that only a minority of English speakers accept. We show that such AdvPs are grammatical as prenominal modifiers, so there is no selectional violation under coordination, and that such AdvPs combine with nouns in regular syntax, rather than via hypothetical compounding. Hence, such constructions do not provide an argument against the symmetric nature of coordination.

Keywords: unlike category coordination, selectional restrictions, adverbs, compounds

1 Introduction

Apparent violations of selectional restrictions, illustrated in (1a) (Sag et al. 1985:165, (124b)), figure prominently in discussions about the structure of coordination.

(1) a. You can depend on [[$_{\mathrm{NP}}$ my assistant] and [$_{\mathrm{CP}}$ that he will be on time]].
b. You can depend on [$_{\mathrm{NP}}$ my assistant].
c. *You can depend on [$_{\mathrm{CP}}$ that he will be on time].

What is interesting about (1a) is that it seems to involve coordination of an NP and a CP in a position in which this CP alone would not be acceptable; see (1c). Such examples are often

We wish to thank the following people for their comments on previous versions of this article and/or for their native judgments: Bob Borsley, Benjamin Bruening, Rui Chaves, Mary Dalrymple, Jakub Filonik, Jamie Findlay, Stephen Jones, Tracy Holloway King, Joseph Lovestrand, John J. Lowe, Jiayi Lu, Berke Şenşekerci, Andrew Spencer, Amanda Thomas, Danfeng Wu, and Sebastian Zawada. We are also grateful to two anonymous reviewers, whose comments led to many substantial improvements. Katarzyna Kuś and Berke Şenşekerci were instrumental in implementing the experiment—funded in part by the National Science Centre, Poland (OPUS-25 project 2023/49/B/HS2/02546 "Coordination: Symmetric or Asymmetric?")—reported in section 4.2. Additionally, Adam Przepiórkowski gratefully acknowledges the Bekker NAWA mobility grant (BPN/BEK/2023/1/00064 "Coordination in Minimalism") to MIT. When preparing and submitting the manuscript, Agnieszka Patejuk was also affiliated at the University of Oxford and was receiving the MNiSW/MEiN scholarship for young outstanding scientists, which she gratefully acknowledges.

Linguistic Inquiry, Early Access Corrected Proof
1–29

https://doi.org/10.1162/ling_a_00548

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

interpreted as providing support for asymmetric theories of coordination, on which a single conjunct determines the category of the coordinate structure (e.g., Munn 1993:79–80, Johannessen 1998:14, Zhang 2009:50–51, 2023:4–5).

While all selectional violation examples in Sag et al. 1985 involve an apparent coordination of an NP and a CP in a strictly nominal position, as in (1), Bruening and Al Khalaf (henceforth B&AK) (2020) argue that there is another kind of selectional violation, involving an apparent coordination of an AdvP and an AdjP in a strictly adjectival prenominal position, illustrated in (2) (B&AK 2020:2, (4)).[1]

(2) a. *The* [[$_{\text{AdvP}}$ *Once*] *and* [$_{\text{AdjP}}$ *Future*]] *King* (book title)
b. *the once king

This second kind of apparent selectional violation has also been interpreted as providing an argument for asymmetric theories of coordination (Zhang 2023:5n2).

However, other cases of unlike category coordination seem to provide evidence for symmetric theories of coordination, on which categories of all conjuncts contribute to the category of the coordination (e.g., Borsley 1994:230–231, 2005:463–465, Dalrymple 2017:34–35, Neeleman et al. 2023:57–58).[2] This is illustrated in (3) from B&AK 2020:25, (85).

(3) a. Danny became [[$_{\text{NP}}$ a political radical] and [$_{\text{AdjP}}$ very antisocial]].
b. *Danny became [[$_{\text{NP}}$ a political radical] and [$_{\text{PP}}$ under suspicion]].

Given that *become* selects for an NP (*Danny became a political radical* is fine) or an AdjP (*Danny became very antisocial* is also fine), but not a PP (**Danny became under suspicion* is bad), [NP & AdjP] is a possible argument of *become* (see (3a)), but a coordination involving a PP conjunct is not (see (3b)). That is, categories of all conjuncts must satisfy selectional restrictions, contrary to predictions of asymmetric theories of coordination. To the best of our knowledge, contrasts such as (3a–b) have not been addressed by the proponents of asymmetric approaches to coordination.

Apparent selectional violation examples such as those in (1) and (2) are also important for linguistic theory at large, as they have been explained using a variety of nonstandard theoretical mechanisms and thus could be construed as providing evidence for such nonstandard mechanisms. For example, the analyses of B&AK and Bruening (2023) assume (a) left-to-right (rather than bottom-up) structure building; (b) categories organized into stacks, with stipulations about which processes only see the top of the stack and which have access to all stack elements; (c) a syntactic mechanism of S(emantic)-selection, whose function is largely the same as that of purely semantic selection, but which has some very special properties crucial in the analysis of coordination; and

[1] The ungrammaticality marking of (2b) is B&AK's, and it will be disputed below.

[2] In this article, we assume that coordination of unlike categories is possible—that is, that the Law of the Coordination of Likes (LCL; Williams 1981:sec. 2), widely assumed until recently, is a myth. Multiple arguments against the LCL may be found in, for example, Levine 2011, Dalrymple 2017, Przepiórkowski 2022, and Neeleman et al. 2023; see also Bayer 1996. While B&AK defend the LCL, this defense is refuted in Patejuk and Przepiórkowski 2023, and—in a reply to that article—Bruening (2023:1) concedes that "[Patejuk and Przepiórkowski (2023)] are correct, and there is no requirement that conjuncts match in syntactic category." Also, while an early draft of Neeleman et al. 2023 assumed the LCL, the published version provides an analysis of coordination of different categories. Hence, there seems to be a growing consensus that unlike category coordination is real.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

so on. Hence, showing that the postulated selectional violations are only apparent would have consequences not only for the theory of coordinate structures, but also for grammatical theory in general.

This is exactly the aim of this article: to show that selectional violations are not real. The apparent "CPs in NP positions" violations (as in (1)) have already been questioned in Patejuk and Przepiórkowski (P&P) 2023 and Kim and Lu 2024. In particular, Kim and Lu (2024) argue on the basis of an acceptability judgment experiment that such apparent violations involve the processing effect of grammaticality illusion (Gibson and Thomas 1999) rather than genuine selectional violations. In the current article, we deal with the purported "AdvPs in AdjP positions" violations (as in (2)) and show—on the basis of corpus and experimental evidence—that they are also not real. Together, the results of these articles remove an important argument for asymmetric theories of coordination and free linguistic theory from the unnecessary burden of the nonstandard mechanisms needed to account for such apparent violations of selectional restrictions in coordination.

2 Empirical Claims

B&AK (2020) claim that it is not possible to coordinate unlike categories (see footnote 2). They account for examples such as *the once and future king*, involving an adverb and an adjective, by analyzing non-*ly* adverbs as adjectives necessarily embedded in adverbial projections: for example, $[_{\text{Adv}}\ [_{\text{Adj}}$ *once*$]\ \emptyset_{\text{Adv}}]$. However, such adverbial projections may be elided in specific configurations in coordination, leaving behind pure adjectives—for example, $\cancel{[_{\text{Adv}}}\ [_{\text{Adj}}$ *once*$]\ \cancel{\emptyset_{\text{Adv}}]}$—resulting in coordination of two adjectives. Hence, all non-*ly* adverbs may in principle be coordinated with adjectives. By contrast, typical adverbs ending in -*ly* are immune to this analysis, so such typical adverbs may not be coordinated with adjectives (e.g., **She became unnerved and distractedly*; B&AK 2020:16, (51b)). Crucially, B&AK claim that the relevant adverbs alone cannot be prenominal modifiers—that is, that NPs such as *the once king*, *the twice president*, and *the soon return* are ungrammatical.

All of B&AK's relevant examples involve just five adverbs: *once*, *twice*, *thrice*, *now*, and *soon.* In P&P 2023, we show that many other non-*ly* adverbs do not behave like *once* and cannot be coordinated with an adjective. There (P&P 2023:342, (72)–(75)), we also present the following attested examples, which demonstrate that *once*, *twice*, *now*, and *soon* can in fact be used as prenominal modifiers on their own, contrary to the purported ungrammaticality of (2b) (*the once king*):

(4) The Once-King of Penn State

(5) Twice Winner of the Man Booker Prize

(6) Many years ago Korn's father had dealings with the now president.

(7) They call him the Thane of Glamis, Thane of Cawdor, and the soon king.

If these examples are grammatical and adverbs such as *once* may be used prenominally, then examples such as (2a) (*the once and future king*) do not involve selectional violations. We conclude

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

in P&P 2023 that the following generalization from Huddleston and Pullum's (2002) *Cambridge Grammar of the English Language* (*CGEL*), which precludes selectional violations, is essentially right:

(8) If (and only if) in a given syntactic construction a constituent X can be replaced without change of function by a constituent Y, then it can also be replaced by a coordination of X and Y.
(Huddleston and Pullum 2002:1323)[3]

In a reply to P&P 2023, Bruening (henceforth B) (2023) agrees with the "if" part of (8), but not with the "only if" part, claiming that the two kinds of apparent selectional violations discussed by B&AK are "fully general" and provide valid counterexamples to (8). However, rather than defending B&AK's analysis of the apparent "AdvPs in AdjP positions" violations, which relies on the -*ly* vs. non-*ly* distinction, B proposes a new analysis, one that relies on idiosyncratic feature composition of particular adverbs: the five considered in B&AK 2020 (*once*, *twice*, *thrice*, *now*, *soon*) as well as possibly three more "identified" in B 2023:16 (*always*, *often*, *seldom*).

B follows B&AK in claiming that such adverbs may be coordinated with adjectives but cannot be used alone as prenominal modifiers. However, while B dismisses (6)–(7) as "completely unacceptable, in my judgment" (B 2023:8) and (5) as "telegraphic" for *Twice the Winner* . . . (rather than *The Twice Winner* . . . , ungrammatical according to B), he no longer claims that strings such as *once king* in (4)[4] are always ungrammatical, as B&AK did. Instead, B posits that a minority of speakers accept them as compounds. By contrast, coordinations such as *the once and future king* are supposed to be grammatical for all speakers as regular syntactic constructions. Hence, a prediction of this analysis is that there should be no correlation between how readily a speaker accepts the hypothetical [$_N$ Adv N] compound (this depends on their lexicon) and how readily they accept the corresponding [Adv & Adj N] syntactic construction (this depends on their grammar).

Our position is that the relevant [Adv N] constructions are not lexical, but syntactic, and in principle grammatical. Speakers may differ in the extent to which they accept specific adverbs as prenominal modifiers, but—given the biconditional in (8)—whether they accept a given adverb as a prenominal modifier or not, they should do so in both syntactic constructions: [Adv N] and [Adv & Adj N]. That is, unlike B, we predict a correlation regarding speakers' acceptance of the two constructions.

3 Corpus Evidence

On the basis of attested data, we first (in section 3.1) show that prenominal adverbs that B (2023) considers unacceptable (*now*, *soon*) may easily be found in corpora and on the Internet, and then

[3] (8) is a variant of "Wasow's Generalization" and together with other such variants it abstracts away from matters of agreement; see Przepiórkowski 2022:secs. 5–6 for discussion.

[4] We follow B in ignoring orthography; that is, we equate *once king* with *Once-King*, and so on.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

(in section 3.2) we refute B's claims about the "telegraphic" nature of strings such as *twice winner*. In section 3.3, we show in detail that constructions such as *once king* are not compounds. These results are consistent with what dictionaries say about *once*, *now*, and so on (see section 3.4). We finish (in section 3.5) with a discussion.

3.1 Now *and* Soon *May Be Used Prenominally*

As noted above, B considers sentences (6)–(7) with prenominal *now* and *soon* completely unacceptable. However, it is easy to find such prenominal adverbs in corpora. For example, in the English Web 2020 corpus,[5] the pattern [*a*/*the* Adv N] produced 12,894 matches for Adv = *now* and 653 for Adv = *soon.* Manual inspection of random samples of 100 matches suggests that over 40% of *now* and over 85% of *soon* matches are true positives, so there should be well over 5,000 genuine instances of [D *now* N] and well over 500 of [D *soon* N].

In fact, following the death of Queen Elizabeth II in September 2022, there was an increased media presence of such prenominal adverbs, especially *now*. Here are a few examples from the BBC (see (9)), *The Guardian* (see (10)), the *Daily Mirror* (see (11)), *Harper's Bazaar* (see (12)), and the *New York Post* (see (13)).[6]

(9) He claims the *now Prince of Wales* ordered him to shave it off after a week-long argument. . . . Harry told Bradby he didn't think the *now Duchess of Sussex* would have recognised him clean-shaven.

(10) At last year's global climate talks in Glasgow, the *now monarch* said: "We know what we must do."

(11) In the letter, the *now king* wrote: "Dear Granny, I am sorry that you are ill."

(12) the *now Prince and Princess of Wales*, William and Catherine

(13) The *once Duke and Duchess of Cambridge* made their first public engagement appearance post-funeral yesterday.

Such adverbs not only occur prenominally alone, but also may occur as second conjuncts in prenominal coordinations.

(14) Umberto II's heir was Prince Vittorio Emanuele, the *then and now* Prince of Naples.

(15) And we share your hope that there is a *final and soon* resolution to the peace process.

Examples such as (14)–(15) directly contradict analyses proposed in B&AK 2020 and B 2023, which predict that adverbs can occur prenominally only when coordinated with an adjective, and then only when the adjective is the last conjunct.

[5] http://www.sketchengine.eu/ (Jakubíček et al. 2013, Kilgarriff et al. 2014).
[6] Specific sources of attested examples are given in the list following the references.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

3.2 Twice Winner *Is Not Telegraphic*

B (2023:8) claims that example (5) in P&P 2023 is "telegraphic. If the determiner were pronounced, it would come after *twice.*" He illustrates this claim with (16) (his (21)).

(16) a. (It is) twice the winner of the Man Booker Prize.
b. *(It is) the twice winner of the Man Booker Prize.

However, contrary to B's ungrammaticality marking, examples such as (16b) are grammatical and easily found in corpora: there are 40 matches for the specific sequence of lexemes *a/the twice winner* in English Web 2020—all true positives—including those in (17)–(19).

(17) He is the *twice* winner of the UK open Kumite Karate Championships.

(18) The Londoners travel to the *twice* winners of the League Cup.

(19) He was also a *twice* winner of the Paris Six Hour Enduro.

Corpora and Google also provide numerous examples of *twice* pre-modifying nouns other than *winner*, including *champion* and *loser*, for instance.

(20) The sledge hockey team Udmurtia is the *twice* champion of Russia.

(21) Ted Griffin followed on as holder and the *twice* loser of the shield.

The same applies to *thrice* (also discussed by B&AK). For instance:

(22) "Old Tom" is the *thrice* winner of the first major golf tournament.

(23) Citadel's billionaire investor Ken Griffin calls Trump a '*thrice* loser'

(24) A *thrice* world and *twice* Pan-American Champion, Bia now has her sights set on the European Jiu-Jitsu Championship.

In summary, both *twice* and *thrice* may in fact act as direct pre-modifiers of nouns, contrary to B's claim about the "telegraphic" nature of *twice winner* in (5).

3.3 Prenominal Adverbs Do Not Form Compounds

B&AK (2020:14–15) state that strings such as *the once king*, *the twice president*, *the soon return*, and *the now Democrat* are ungrammatical. This claim cannot be maintained given the attested examples in P&P 2023 and B's (2023) own acceptability judgment experiment (discussed in section 4 below), which shows that a considerable number of native speakers accept such strings. In order to retain the analysis of constructions such as (2a) (*the once and future king*) as involving selectional violations, B posits that the relevant adverbs combine with nouns not via regular syntactic processes, but via compound-forming processes; for example, *once king* is supposed to have the lexical structure [$_N$ *once king*]. On this view, the relevant adverbs cannot directly combine with nouns in syntax, but they can still be initial conjuncts in prenominal coordinations, in an apparent selectional violation.

If B were right, this would be an important discovery of a new kind of compound in English, not registered by linguists specializing in compounding so far, adding [Adv N] to the repertoire

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

that already contains [Adj N] (e.g., *blackbird*) and [N N] (e.g., *bird flu*).[7] However, B's main argument for this claim is based on the anaphoric *one.* B assumes that *one* cannot refer to the noun within an [Adv N] compound.[8]

In an acceptability judgment experiment, B considers pairs such as (25a–b).

(25) a. We want to meet the now Caliph and the old one. (Adv)
b. We want to meet the current Caliph and the old one. (Adj)

On the compound hypothesis, (25a) is supposed to be ungrammatical, as *one* attempts to refer to *Caliph*—a proper part of the hypothetical compound [$_N$ *now Caliph*]. On the other hand, (25b) is grammatical because *current Caliph* is a regular syntactic construction with a syntactic constituent, [$_N$ *Caliph*], to which *one* may refer. In section 4, we show that the results of this experiment do not support B's compound hypothesis; here, we demonstrate that corpus data also contradict it.

[Adv N] constructions are relatively rare and *one*-anaphora is also textually infrequent, which makes it difficult to find examples, similar to (25a), which would combine both these rare phenomena. Nevertheless, (26)–(27) are two such attested examples; in both, *one* refers to the noun in *now wife.*

(26) In his will he says 'My *now* wife,' as though he had had a previous *one.*

(27) Boris Johnson . . . Private Eye reports he was caught receiving a blow job in the Foreign Office when he was Foreign Secretary by his *now*-wife, while he was still married to the previous *one.*

However, the *one*-anaphora test is only one of many tests employed to distinguish compounds from regular syntactic constructions—all imperfect. It is routinely noted that "there are (almost) no reliable criteria for distinguishing compounds from phrases" (Lieber and Štekauer 2009a:14). As discussed in Giegerich 2009, there are two criteria that are commonly assumed to be valid, but in one direction only: a given construction is a compound if (but not only if) it satisfies at least one of them. The first is phonetic: when the first element is stressed, as in *blackbird* (a particular species), but not in *black bird* (any bird that is black), it is a compound, not a phrase. The second is semantic: the meaning of *blackbird* is not fully compositionally derived from *black* and *bird* (it is not just any bird that is black), so it is a compound, not a phrase. These tests do not provide arguments for the compound status of *once king* and the like, as the stress is on the noun and the meaning is fully compositional, but they also do not exclude this possibility.

[7] See, for example, Lieber 2009 and Bauer 2020 on types of compounds in English.

[8] B does not say anything about the nature of compound formation. One way of substantiating B's analysis is to assume that compounds are complex words formed within the lexicon and that words in general are boundaries for *one*-anaphora. However, elsewhere (B 2018a) he argues that compounding takes place in syntax and that compounds are not boundaries for *one*, citing example (i) (his (12c); brackets indicating compound boundaries and italics added).

(i) The old [the-*dog*-ate-my-homework excuse] won't work because I know you don't have *one*!

If so, additional assumptions would be needed to prevent *one* from referring to the noun in hypothetical [Adv N] compounds.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

As for syntactic tests, the *one*-test is used relatively frequently, but—as discussed, for example, in Giegerich 2009:sec. 9.5—it is known to be unreliable. A test that works better, in the sense that apparent counterexamples are easier to account for, is the internal modifiability test: if the first or the second element may be modified, then it is not a compound. Lieber and Štekauer (2009a:11–12) also consider modifiability of the second element to be the most reliable test, followed by modifiability of the first element (with *one*-anaphora as a third, less reliable test).[9] According to the first of these tests, *black bird* is not a compound, as *black ugly bird* is fine. By contrast, *ugly* cannot modify *bird* alone in *blackbird*, so *blackbird* is a compound. According to the second test, *very black bird* is fine, so *black bird* is not a compound. By contrast, a similar modification is impossible in the case of *blackbird*, without losing its reference to the particular species.

It is possible to demonstrate that, according to these more reliable tests, the relevant [Adv N] constructions are syntactic phrases, not compounds. Considering the first test, we first looked in English Web 2020 for constructions such as *a/the twice/thrice . . . winner/champion*, where . . . stands for 1 to 5 words. Among 169 hits, many involved the name of an award or competition: for example, *the twice Booker Prize winner*, *the twice Paris Show champion.* We excluded them, as it could be claimed that *Booker Prize winner*, *Paris Show champion*, and the like are complex compounds, to which *twice* attaches via lexical processes, resulting in even more complex compounds. In the following examples, such a complex compound analysis is unlikely:

(28) Maghsoodloo is a *twice* [Iranian national champion].

(29) Paikidze . . . is a *twice* [U.S. women's champion].

(30) Zhang Zhong . . . is a Chinese chess grandmaster, a *twice* [Chinese champion].

According to B, all these examples should be ungrammatical or only have the reading where the adverb modifies the immediately following adjective; the following examples, ungrammaticality markings, and comments are B's (2023:26, (66b), (67a–b)):[10]

(31) *the once beloved king (only grammatical on parse *the once-beloved king*)

(32) *the twice happy winner (only grammatical on parse *the twice-happy winner*)

(33) *a soon short visit (only grammatical on parse *a soon-short visit*)

On this analysis, Maghsoodloo in (28) would be a champion who is *twice-Iranian* (whatever that means), Paikidze in (29) would be *twice-U.S.*, and Zhang Zhong in (30) would be *twice-Chinese.*[11]

[9] *CGEL* (p. 449) also relies on such modifiability tests, and not on *one*-anaphora.

[10] Note that, while the interpretation of (31) as 'the king who was once beloved' is coherent, but not the only reading, the interpretation of (32) as 'the winner who was happy twice' is clearly not the intended reading, and the interpretation of (33) as 'a visit that will soon be short' is borderline incoherent.

[11] We are not denying the *possibility* that the sequence Adv Adj N may have the structure [Adv Adj] N, but examples (28)–(30) demonstrate that it may also have the structure Adv [Adj N]. This structural ambiguity is clear in the following constructed example:

(i) The *once popular mayor of London* became . . .
 a. . . . the most unpopular mayor in England. (assumes [*once popular*] *mayor* . . .)
 b. . . . an equally popular prime minister. (assumes *once* [*popular mayor* . . .])

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

A very different example of modification of the second element that contradicts B's compound hypothesis is (34).

(34) The *now* [alleged scammer and convicted felon], Danielle Miller, was a 13 year old girl, who sent her crush videos of herself masturbating with a Swiffer mop.

This example is particularly interesting, as—on the most prominent interpretation—*now* combines with the coordination *alleged scammer and convicted felon*, clearly a syntactic construction. Another example involving coordination is (35).[12]

(35) After the *soon* [prime minister and alternate prime minister designates] concluded their remarks . . .

The relevant adverbs may also modify nominal constituents containing relative clauses—that is, again, uncontroversial syntactic (not lexical) constructions.

(36) The *once* [mother who was so caring] is no longer caring to her family members.

(37) The *once* [mother who had all the time in the world for her children] suddenly began spending all her kid's trust fund.

(36) cannot involve the structure "[once mother] who was so caring," with the relative clause modifying the hypothetical compound *once mother.* This is clear from the context: (36) is taken from a forum, where the poster complains that her mother suffers from depression and is not caring toward her family anymore. That is, (36) is about somebody who is still a mother, contrary to the compound analysis, which would be about somebody who was a mother once. (37) is analogous, with an even larger syntactic construction.

Similarly, (38) is about Bill Hudson, who disowned two of his five children; that is, it is about a father (just, in a sense, not of five anymore) rather than about a *once father.*

(38) "I no longer recognize Oliver and Kate as my own," the *once* [father of five] told Sunday's *The Mail*.

Considering the second test, we looked for examples of *very* modifying the adverb in [Adv N]. This is in principle only possible in the case of the qualitative adverbs *soon*, *seldom*, and *often* (cf. **very once*, **very twice/thrice*, **very now*, **very always*), so we looked for *a/the very soon/seldom/often* followed by a noun. (39)–(44) are just 6 of the 59 examples found in English Web 2020.[13]

(39) I wish nothing but label troubles, bankruptcy, and a *very soon* breakup for Deathstars in the future.

(40) [T]his co-evolution is the best way to assure a *very soon* death for VB.NET.

[12] The possibility of such coordinations is sometimes treated as a separate argument against compoundhood; see, for example, *CGEL* (pp. 449–451).

[13] Some of the native speakers we consulted found these examples—especially (41)–(44), involving *seldom* and *often*—markedly worse than many of the other attested examples cited in this article.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

(41) Violence and street fights occur on a *very seldom* basis.

(42) It seemed to be one of the *very seldom* bugs in the game.

(43) They are a *very often* occurrence at our office, she said.

(44) [T]hey realize about the *very often* appearance of a sola flower bouquet in arts.

We conclude that the two most reliable syntactic tests for compoundhood show unequivocally that the relevant [Adv N] constructions are syntactic, not lexical. This—together with the *now wife . . . previous one* examples (26)–(27), which show that such constructions are syntactic according to B's own criteria as well, and with the correct interpretation of the results of B's experiments offered in section 4—leaves no doubt that the relevant [Adv N] constructions cannot be analyzed as compounds. In what follows, we demonstrate that this conclusion is compatible with what dictionaries say about *once*, *now*, and so on.

3.4 Dictionaries

B&AK (2020) assume that the relevant adverbs cannot directly pre-modify nouns, and B (2023) claims that even if sometimes they can, it is only on a compound analysis. Such views should be contrasted with the fact that the *Oxford English Dictionary* (*OED*) records prenominal uses of the relevant adverbs and analyzes most of them in such uses as regular adjectives (*once*, *now*, *soon*, *seldom*—marked as "now chiefly U.S.," *often*—marked as "now rare") or as "quasi-adjectives"[14] (*twice*, *thrice*)—crucially, as syntactic modifiers, not as parts of compounds. Some of the examples cited in the *OED* in such lexical entries are given in (45)–(51).

(45) the *once* queen who liked to play at milkmaid

(46) the *twice* pilgrim

(47) till the *thrice* Confession Blot the thrice Denial out

(48) my *now* wife

(49) a white mist, thick, in the *soon*-twilight to be impenetrable

(50) with her small *seldom* smile

(51) a frequent mingling and an *often* hospitality

Some of the relevant adverbs also have adjectival entries in other dictionaries. For instance, the Internet edition of *Merriam-Webster* (https://www.merriam-webster.com/dictionary/) lists adjectival *once* (exemplified with (52)), *now* (with (53)),[15] and *seldom* (no example); Dictionary.com recognizes adjectival *once*, *seldom*, and—marked as "archaic"—*often*; and so on.

[14] The term *quasi-adjective* is used in some unrevised *OED* entries, while in revised entries "this term is not used: a word behaving as an adjective . . . is treated as an adjective" (https://public.oed.com/how-to-use-the-oed/glossary-grammatical-terms/).

[15] Recall that (6), containing the sequence in (53), was "completely unacceptable, in my judgment" for B.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

(52) contributions to enrich the legal resources of the *once* province of Britain

(53) the *now* president

It seems clear that dictionaries classify the relevant adverbs as adjectives on the basis of the usual definition of English adjectives as "words that modify nouns" (see Payne, Huddleston, and Pullum 2010 for discussion). Whether they are right or not, the fact that such prenominal uses of these adverbs have long been recorded in dictionaries confirms our analysis of the relevant constructions as syntactic and hence supports our rejection of selectional violations in coordination involving hypothetical "AdvPs in AdjP positions."

3.5 Discussion

On the basis of rich attested data, we demonstrated that there are no "AdvPs in AdjP positions" selectional violations in coordination: both configurations involving the relevant apparently adverbial words—[Adv & Adj N] and [Adv N]—are grammatical as syntactic constructions. The correct generalization seems to be that temporal and frequentative non-*ly* adverbs may pre-modify nouns (as in (54a), repeated from (53)) and hence may be coordinated with other such pre-modifiers, including adjectives (as in (54b)), conforming to *CGEL*'s view of coordination in (8).

(54) a. the *now* president
b. the *now* and future president

We see at least two ways of modeling this generalization. The first, consonant with the discussion in section 3.4, is that such temporal and frequentative words are in general categorially ambiguous between adverbs and adjectives. On this conservative view, *now* in (54a–b) is an adjective, so (54b) involves coordination of two adjectives; hence, there is no selectional violation of any kind. The second possibility is that pre-modifiers of nouns constitute a more heterogeneous class that consists not only of adjectives, but also of the relevant adverbs, some prepositions (Payne, Huddleston, and Pullum 2010:40), and perhaps nouns.[16] On this view, different categories are coordinated in (54b), but—given the grammaticality of (54a)—also without any selectional violations.

No matter how such facts are analyzed, the generalization in terms of the semantically coherent class of temporal and frequentative adverbs seems to be robust, as other adverbs in this class behave similarly. This is most clear in the case of *then*, which—just like the other relevant adverbs—may be used both in [Adv & Adj N] and in [Adv N] configurations, as attested by the following examples:[17]

(55) a. There he met some of the *then and future* leaders of Fiji.
b. Sergey Bogdanchikov, the *then and current* President of Rosneft

[16] Not all [N N] constructions are reasonably analyzed as compounds; see, for example, *CGEL* (chap. 5, sec. 14.4), Giegerich 2009, Lieber and Štekauer 2009a, and references therein.

[17] There are over 100,000 occurrences of the [D *then* N] pattern in English Web 2020, including (56), which makes *then* a particularly productive prenominal adverb.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

(56) a. Lines were crossed, in particular by the *then* Director of the FBI.
b. However, the *then* Prime Minister, Bulent Ecevit, rejected the offer.

Such "adjectival" uses of *then* are also recorded and exemplified in the *OED* and *Merriam-Webster.* A similar case could be made for *yesterday* (also recorded as an adjective in the *OED* and Dictionary.com) and perhaps also for *tomorrow*, *ever* (recorded in the *OED* as "obsolete" and in Dictionary.com as "South Midland and Southern U.S."), and so on for various other adverbs in various other dictionaries.

4 Experimental Evidence

In the previous section, we showed that the adverbs considered by B&AK (2020) and B (2023) may be used as direct pre-modifiers of nouns and that such uses should not be analyzed as involving compounding. Here, we first (section 4.1) discuss problems with B's two acceptability experiments, then (section 4.2) present a new experiment, whose results directly contradict B's empirical claims, and finally (section 4.3) conclude with a discussion.

4.1 B's Experiments

The aim of B's (2023) experiments was to show that *syntactic* constructions such as *the now president* are ungrammatical; if such strings are acceptable, it is only on the compound interpretation, where *now president* is a noun with internal *lexical* structure. In this section, we point out some problems with B's interpretation of the results of these experiments. The conclusion is that these experiments do not provide any evidence for the compound analysis of [Adv N].

4.1.1 Problem 1: Interpreting Lower Naturalness as Ungrammaticality In B's first experiment, stimuli included minimal pairs differing only in the use of an adverb or an adjective as a prenominal modifier, as in the following pair (B 2023:tab. 1):

(57) a. We want to meet the now vice president. (Adv)
b. We want to meet the current vice president. (Adj)

In that experiment, performed via Amazon Mechanical Turk, in which responses of 65 native speakers of English recruited from within the United States were taken into account, eight such pairs were used, two for each of *now*, *soon*, *once*, and *twice.*

Participants were asked to evaluate stimuli on a 5-point Likert scale, where the five points were labeled as follows: 1: *Extremely unnatural*, 2: *Somewhat unnatural*, 3: *Possible*, 4: *Somewhat natural*, 5: *Extremely natural.* These names are used by Gibson, Piantadosi, and Fedorenko (2011), whose methodology B followed. They are appropriate for making conclusions about one construction being less natural than another construction. However, scores resulting from such a scale cannot be directly mapped into "grammaticality" and "ungrammaticality."

In particular, nothing justifies translating only average scores of 4 or above to "grammatical" and all scores below 3 to "ungrammatical," and this is how these scores are interpreted in B 2023. This is unjustified because grammatical constructions that are pragmatically deviant or have more

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

Table 1
Results of Bruening's (2023) first experiment

	Grammatical fillers	Ungrammatical fillers	Adjective	Adverb
Mean	4.45	2.27	4.75	3.16
SD	0.91	1.36	0.55	1.27

Source: Bruening 2023:10

natural or frequent paraphrases may easily be judged as *Possible* (score 3) or even *Somewhat unnatural* (score 2).[18]

The mean scores for versions with an adjective and with an adverb, as well as for grammatical and ungrammatical fillers, are given in table 1.[19] About these results, B (2023:11) says, "14 of 65 [participants] rated Adv items 4 or higher (22%), while 28 of 65 rated Adv items below 3 (43%). . . . This means that approximately 43% of those surveyed find non-*ly* adverbs unacceptable as prenominal modifiers. However, approximately 20% rated Adv items 4 or higher, which would seem to indicate that approximately 20% of native English speakers accept these non-*ly* adverbs as prenominal modifiers." Note the lack of balance in this interpretation: averages in the half-open interval [1, 3) of length 2 (see "below 3" in the quotation) provided by 43% of respondents are considered as indicating unacceptability, while only those in the closed interval [4, 5] of length 1 (see "4 or higher") provided by 22% of respondents are considered acceptable. This excludes from consideration 35% of respondents, whose average scores were in the interval [3, 4). We claim that such scores could also reasonably be interpreted as indicating acceptability.

Given the setup of the experiment, each respondent saw four sentences containing an Adv. Assume that three sentences were judged as *Possible* (i.e., 3) and one as *Somewhat unnatural* (i.e., 2). This gives the average score of 2.75—below 3, yet indicating that, for this respondent, the construction with Adv is *possible*, even if occasionally *somewhat unnatural.* A reasonable interpretation would be that, for such a speaker, the Adv constructions are grammatical, even if dispreferred with respect to the usual Adj constructions.

So, given how the points on the scale are named, the opposite imbalance would be much more justified: treating responses averaging between *Possible* and *Extremely natural* as acceptable, and only those below *Somewhat unnatural* as unacceptable. But then, not just 22% but as many

[18] That is, we side here with Bruening 2018b: "It is not possible to decide that below a given mean rating (say, 2.5) sentences are ungrammatical, and above that they are grammatical. For one thing, the judgments reported by subjects are judgments of acceptability, not grammaticality, and judgments of acceptability are affected by numerous non-grammatical factors (such as length, complexity, and familiarity)."

[19] *SD* stands for *standard deviation.* Strictly speaking, it is not appropriate to report means and standard deviations for ordinal data (such as a Likert scale, especially one with named points) or to train linear models on such data (see sections 4.1.2–4.1.3), but this is a relatively common practice in experimental syntax.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

as 22% + 35% = 57% of respondents would have to be interpreted as accepting the relevant [Adv N] constructions.[20]

4.1.2 Problem 2: Ignoring Additive Effects The existence of a considerable number of speakers who accept [Adv N] constructions (e.g., *the once president*) means that the textual presence of [Adv & Adj N] constructions (e.g., *the once and current president*) cannot be immediately interpreted as "AdvPs in AdjP positions" selectional violations: perhaps such [Adv & Adj N] examples were produced by speakers who accept [Adv N] constructions in the first place. In an attempt to defend his analysis of apparent selectional violations, B claims that speakers only accept [Adv N] constructions on the compound reading of [$_N$ Adv N]. In the case of the example at hand, this means that *once president* must be treated as a compound, that is, a single word.

As discussed in section 3.3, B uses the *one*-substitution test to distinguish compounds from syntactic constructions. This test is based on the assumption that *one* may refer only to syntactic items, not to proper parts of compounds. On this assumption, (58a) should be ungrammatical, as *one* attempts to refer to *president*, which is a part of the purported compound *once president*, while (58b) is grammatical, as *one* refers to the syntactic item *president* (*past president* is not a compound).

(58) a. At the event we saw the once president and the current one. (Adv)
b. At the event we saw the past president and the current one. (Adj)

(58) is one of eight minimal pairs (see (25) for another one) used in B's second experiment. This experiment had a design analogous to that of the experiment reported above (with the same names of the 5-point Likert scale) and used responses of 77 participants. The results of this experiment are presented in table 2; for comparison, the results from the first experiment given in table 1 are repeated in table 2 in parentheses (in gray). B (2023:12) summarizes these results thus:

Table 2
Results of Bruening's (2023) second experiment (with comparisons to the first experiment in gray)

	Grammatical fillers	Ungrammatical fillers	Adjective	Adverb
Mean	4.39 (vs. 4.45)	2.20 (vs. 2.27)	3.82 (vs. 4.75)	2.60 (vs. 3.16)
SD	0.95 (vs. 0.91)	1.27 (vs. 1.36)	1.08 (vs. 0.55)	1.17 (vs. 1.27)

Source: Bruening 2023:12

[20] A related but distinct problem is that B seems to assume that all grammatical sentences should have the same high acceptability. This is evident in the following quotation: "If P&P are correct and these non-*ly* adverbs are capable of being prenominal modifiers, then we should not see any difference in the ratings participants assign to the two members of each pair" (B 2023:9). A similar misinterpretation of P&P's claims may be found in the following quotation: "This indicates that, contrary to what P&P say, non-*ly* adverbs are not rated the same way as adjectives in prenominal position" (Bruening 2023:13). Experimental literature is teeming with reports of uncontroversially grammatical constructions differing in acceptability in ways that are statistically significant; see, for example, Francis 2022 and various articles in Goodall 2021 for overviews and examples. So, contrary to these quotations, the decreased acceptability of [Adv N] in comparison to [Adj N] does not automatically mean that [Adv N] is ungrammatical. On the other hand, this difference in acceptability does call for an explanation, and we offer some thoughts on this matter in section 4.3.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

"Participants behaved as expected on the grammatical and ungrammatical fillers. The Adj items were rated slightly lower than the grammatical fillers, but still close to 4. The crucial items are the Adv items. These were rated very low, below 3, like the ungrammatical fillers."

We believe that the crucial items here are not only the Adv items, but also the Adj items. B does not comment on perhaps the most conspicuous difference between the results of the two experiments: the sharp drop in acceptability of Adj constructions from 4.75 in the first experiment to 3.82 in the second experiment (with the accompanying rise of the standard deviation from 0.55 to 1.08).[21]

What may be observed here is a prototypical case of a well-known phenomenon in experimental syntax, namely, the (under)additive effect of multiple conditions (see, e.g., Keller 2000 and Hofmeister, Staum Casasanto, and Sag 2014, as well as Francis 2022 for an overview). It is true that the first experiment shows that the relevant [Adv N] constructions are less acceptable (in the sense of being rated as less natural) than corresponding [Adj N] constructions. However, it is not clear whether the second experiment adds anything to this observation: [Adv N] constructions could have been rated lower in this experiment than in the first one simply because sentences involving the complex construction with *one*-anaphora are *generally* rated lower than the much simpler stimuli used in the first experiment.

In order to test this possibility, it is necessary to replace the two monofactorial analyses of the two experiments reported in B 2023 with a single multifactorial analysis, not reported there. We report on such an analysis in the following section.

4.1.3 Multifactorial Analysis Considered individually, the models used in B's two experiments, developed using the tools made available by Gibson, Piantadosi, and Fedorenko (2011), are state-of-the-art (but see footnote 19): they are mixed-effects linear models, fitted with the R (R Core Team 2012) package lme4 (Bates et al. 2015), with speakers and test items treated as random effects, and the Adv/Adj condition as the single fixed factor. The problem is that the results of such two separate monofactorial models cannot be directly compared in a statistically reliable way.

For this reason, we used the raw results of B's experiments and fitted a 2×2 bifactorial model.[22] The two fixed factors were part of speech (Adv vs. Adj, as in both experiments) and construction (simple in the first experiment and complex *one*-anaphora construction in the second experiment). We followed B in using mixed-effects linear models fitted with the R package lme4, with speakers and test items treated as random effects.[23] Also following B, *p*-values were extracted from the fitted model using the Satterthwaite approximation implemented in the lmerTest package (Kuznetsova, Brockhoff, and Christensen 2017).

[21] Note that, assuming B's interpretation of the results of the first experiment, where only average scores of 4 and higher were considered to indicate acceptability, such uncontroversially grammatical constructions with adjectives should not be considered acceptable, as $3.82 < 4$.

[22] We are grateful to Benjamin Bruening for making the raw results of his experiments available to us.

[23] The maximal model that converged is given by the formula Score $\sim 1 + \text{pos} * \text{construction} + (1 + \text{pos} \mid \text{Participant}) + (1 + \text{pos} \mid \text{Input})$, where *Score* is the expected score on the 1–5 Likert scale, *construction* is either simple or with *one*-anaphora, *pos* is Adj or Adv, and *Participant* and *Input* are the two random effects.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

The analysis confirms that the acceptability of [Adv N] is highly significantly lower than that of [Adj N] (drop of 1.6; $df = 32.3$, $t = -7.50$, $p < 0.001$), but also that *one*-anaphora causes a highly significant decrease in acceptability over the simple construction (drop of 0.9; $df = 19.7$, $t = -4.82$, $p < 0.001$). Importantly, the interaction of these two conditions is slightly underadditive: the co-occurrence of [Adv N] and *one* increases acceptability (by 0.4) with respect to what might be expected if the effect of the two conditions were fully additive, although this interaction is not statistically significant ($df = 30.4$, $t = 1.4$, $p = 0.16$). Hence, the low rating of Adv items in B's second experiment may be explained by the usual (under)additive effect of two independent conditions, hence provides no evidence for B's analysis of [Adv N] as compounds.[24]

4.2 *New Experiment*

B's (2023) analysis of the purported "AdvPs in AdjP positions" violations makes several predictions that his experiments did not test. First, [Adv & Adj N] constructions should be significantly more acceptable than [Adv N] on average, as the former are grammatical for all speakers, while the latter are assumed to be grammatical only for a minority of speakers. Second, the distribution of judgments on [Adv & Adj N] sentences should be unimodal, with the mode comparable to that of grammatical fillers, while in the case of [Adv N] sentences the distribution should be bimodal, with the two modes comparable to those of grammatical and ungrammatical fillers, given the two hypothetical populations of speakers of English differing in whether they have relevant [$_N$ Adv N] compounds in their dictionaries. Third, there should be no correlation between a speaker's acceptance of the *syntactic* construction [Adv & Adj N] and their acceptance of the corresponding apparently *lexical* construction [Adj N].

We tested these predictions in an acceptability judgment experiment involving minimal pairs such as (59a–b).[25]

(59) a. Emily Kuhn spoke with Preston, the now head of global finances at Nestle.
b. Emily Kuhn spoke with Preston, the now and future head of global finances at Nestle.

We considered the same four adverbs that were tested in B's experiments. However, unlike B, we tested each of these adverbs separately and with a specific noun, as compounding is an idiosyncratic process; for example, a speaker might have the hypothetical compound *now head*,

[24] As an anonymous reviewer points out to us, B's second experiment should have had one more condition, testing whether respondents accept simple [Adv N] constructions. Then, considering only speakers whose acceptance of [Adv N] and [Adj N] is comparable, a significantly larger drop in acceptability of *one*-anaphora with [Adv N] items compared to *one*-anaphora with [Adj N] items could be interpreted as an argument for the compound analysis of [Adv N].

[25] The importance of such a direct comparison is acknowledged in B 2023:13: "Importantly for this article, although most English speakers reject non-*ly* adverbs as prenominal modifiers by themselves, they do accept them conjoined with an adjective in this position." However, we cannot agree with B's justification of this claimed acceptability difference given in the next sentence: "While examples like this were not tested directly in the two experiments reported here, B& AK present extensive corpus and judgment data to substantiate this claim." B&AK provide about 15 attested examples of [Adv & Adj N] (and it is equally easy to find numerous attested examples of [Adv N]; around 30 are given in section 3), and they only provide their own judgments on some of the corresponding [Adv N] items.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

Table 3
Expressions tested in the new experiment

Adverb	[Adv N]	[Adv & Adj N]
once	once boss	once and current boss
twice	twice winner	twice or three-time winner
now	now head	now and future head
soon	soon appearance	soon and certain appearance

but not *now prime minister* or *once head.* Table 3 presents the specific sequences tested; they were selected as plausible on the basis of their occurrence in corpora. For each adverb, eight minimal pairs such as (59a–b) were tested. The experiment had the usual Latin square design, so that each respondent saw only one of the two sentences. For technical reasons, the experiment was split into two subexperiments, one containing *once* and *soon* stimuli, the other containing *twice* and *now*; that is, each subexperiment contained 16 stimuli. As this experiment was combined with another experiment, there was an unusually large number of fillers in each subexperiment, namely, 80: 32 test items of that other experiment and 48 true fillers ranging from fully grammatical to fully ungrammatical (but interpretable; i.e., no word salad).[26]

In order to better judge the absolute acceptability, we included among the fillers the standard items from Gerbrich, Schreier, and Featherston 2019. Standard items are sentences grouped into five classes, A–E, known to reliably produce the whole range of different acceptability ratings.[27] Each class consists of three sentences; those in classes A and B are meant to be grammatical, those in C are borderline, and those in D and E are meant to be ungrammatical; see online appendix B for the complete list (https://doi.org/10.1162/ling_a_00548). By comparing ratings of stimuli with the ratings of such standard items, it is possible to assign specific acceptability labels to stimuli.

We measured acceptability using the thermometer method (TM; Featherston 2008, 2009), rather than the more usual Likert scale; as argued in Featherston 2009, the TM combines the strengths of the Likert scale and magnitude estimation methodologies (Bard, Robertson, and Sorace 1996), while avoiding their weaknesses. The idea of the TM is that respondents are presented with two sentences, visible throughout the experiment: one that is relatively unacceptable (but interpretable) and one that is relatively acceptable (but not very natural). To this end, we used the following two sentences from Featherston 2021:53:

(60) The father fetches the wholemeal bread them.

(61) The father fetches the children the wholemeal bread.

[26] The stimuli and fillers, as well as R code and analyses, are available in the following Open Science Framework repository: https://osf.io/j27pn/.

[27] For example, on the 1–5 Likert scale, examples in class A produce average ratings close to 5, B – around 4, C – a little above 3, D – a little above 2, and E – between 1 and 1.5 (Gerbrich, Schreier, and Featherston 2019:314–315; cf. Molimpakis 2019:128–130, Featherston 2020:169–171, Brown et al. 2021:16–18).

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

Sentence (60) is assigned value 20 and (61) is assigned 30, and respondents are asked to judge other sentences by comparing them with these two sentences.[28]

The experiment was implemented in LimeSurvey (https://www.limesurvey.org). Participants were recruited via Prolific (https://www.prolific.com) and remunerated according to its recommendations. A total of 89 native speakers of English completed the experiment, but 19 were excluded from analysis because they did not satisfy one or more of the quality control criteria listed in online appendix B. Of the remaining 70 participants, 24 were female and 46 male, all at least 20 years old (M = 39.7, *SD* = 13.1), all native speakers of English located in the UK (*n* = 56), the US (*n* = 13), or Canada (*n* = 1).

Average scores on [Adv N] and [Adv & Adj N] constructions involving particular adverbs are presented in figure 1. The first prediction of B's analysis, that [Adv & Adj N] should be more acceptable on average than [Adv N], is only borne out in the case of *soon*; the acceptability of [Adv N] is significantly greater than that of [Adv & Adj N] in the case of *once*, and insignificantly so in the case of *twice* and *now* (see the confidence intervals).

Moving to the second prediction, that judgments of [Adv N] should have a bimodal distribution, we utilized three different methods of modality testing, as no single method is generally accepted as fully reliable, namely, methods implemented in the following R packages: LaplacesDemon (Statisticat and LLC 2021), multimode (Ameijeiras-Alonso, Crujeiras, and Rodríguez-Casal 2021; using the default ACR method), and diptest (Mächler 2024). The results are presented in figures 2–5, which contain juxtaposed histograms and density plots of the two constructions,

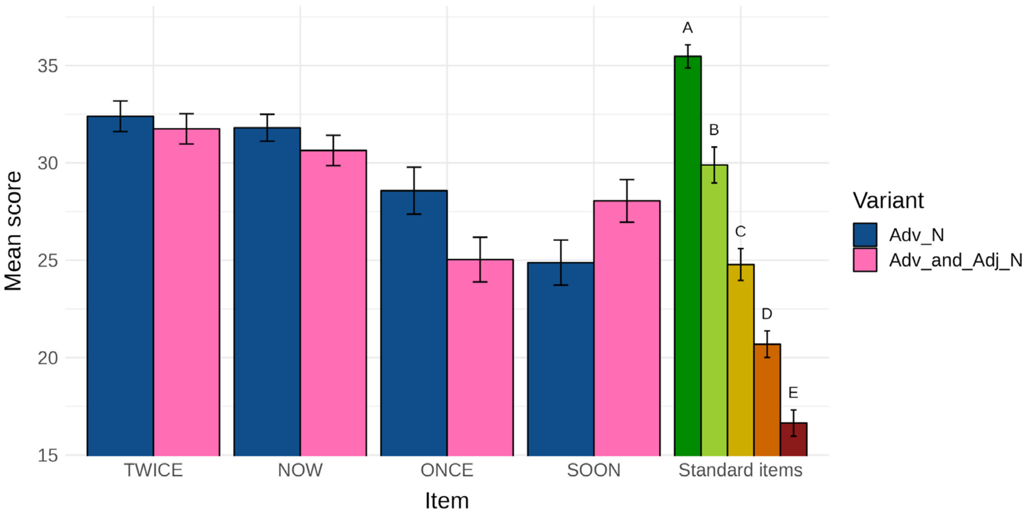


Figure 1
Average scores on stimuli and standard items (with 95% confidence intervals)

[28] Given that the vast majority of judgments in TM studies fall within the range of 15–35 (Featherston 2009:64), the response scale was technically limited to 10–40.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

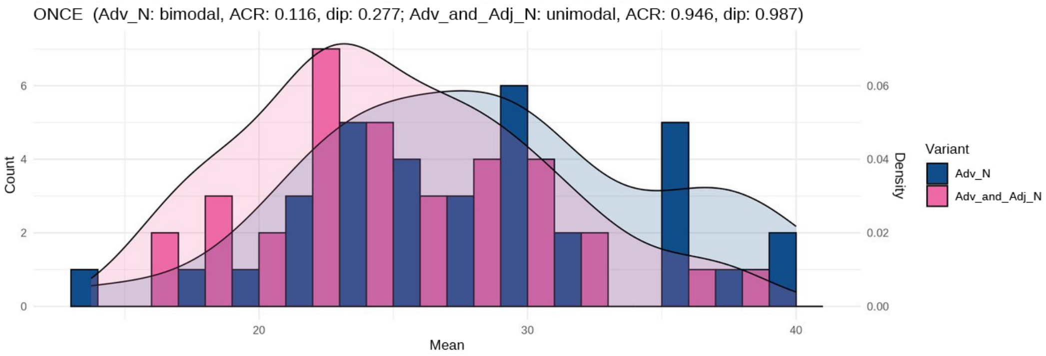


Figure 2
Histograms, density plots, and results of modality tests for *once boss* and *once and current boss*

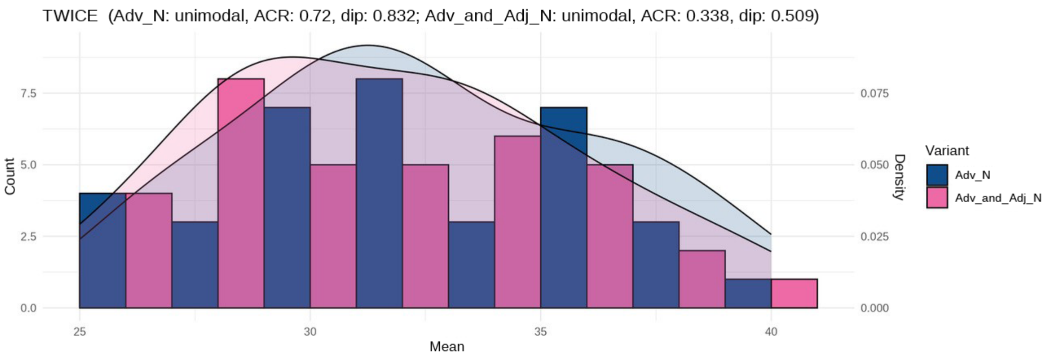


Figure 3
Histograms, density plots, and results of modality tests for *twice winner* and *twice or three-time winner*

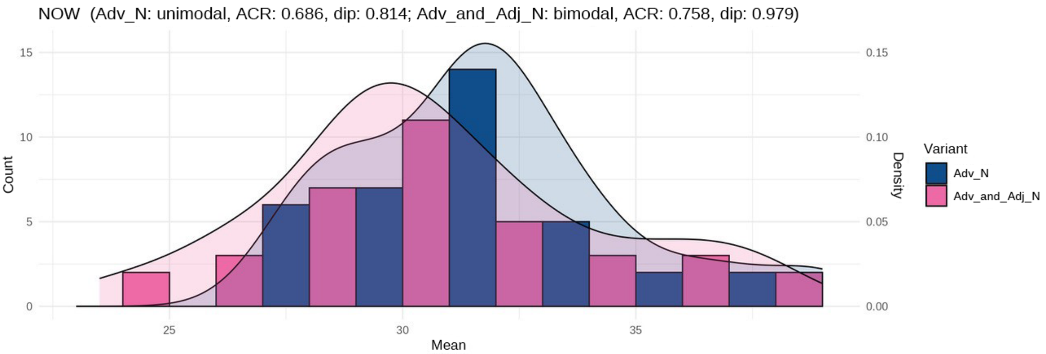


Figure 4
Histograms, density plots, and results of modality tests for *now head* and *now and future head*

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

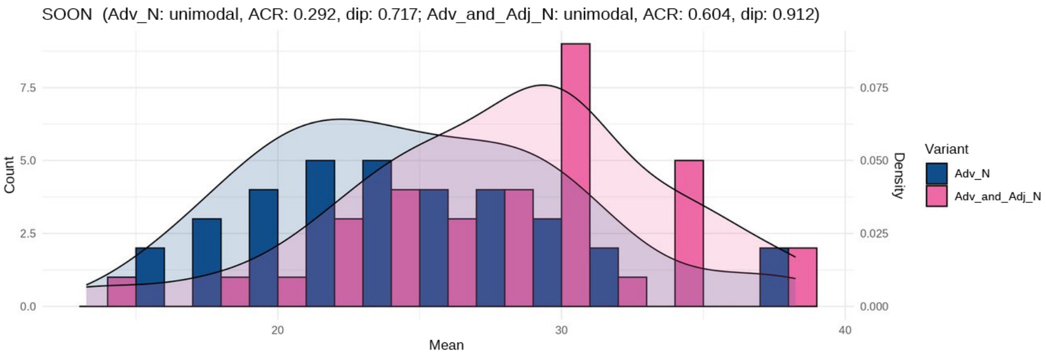


Figure 5
Histograms, density plots, and results of modality tests for *soon appearance* and *soon and certain appearance*

[Adv N] and [Adv & Adj N], for the four adverbs. Titles of these figures contain information about the results of modality tests for both constructions. First, *unimodal* vs. *bimodal* reports the results of is.unimodal and is.bimodal functions from the LaplacesDemon R package. The following ACR and dip numbers are *p*-values reported by functions modetest (from the multimode package) and dip.test (from the diptest package). Values below 0.05 would make it possible to reject the null hypothesis that the distribution is unimodal and claim that it is multimodal.

Note first that all reported *p*-values are considerably greater than 0.05, so ACR and dip did not find multimodality in any of the datasets. LaplacesDemon also reports unimodality for three out of four [Adv N] constructions, apart from *once boss.* However, instead of the expected modes corresponding to unacceptability (i.e., around 20 or less) and acceptability (around 30 or more), the detected modes are 27.6 and 36.6, which—by comparison to the standard items—may be interpreted as "marginal acceptability" and "extreme acceptability."[29] Hence, it is safe to conclude that judgments on both constructions have unimodal distributions, contrary to B's analysis, which predicts bimodality in the case of [Adv N].[30]

Finally, if [Adv N] constructions were lexical and [Adv & Adj N] syntactic, as in B's analysis, there should be no correlation between their levels of acceptability. We tested this third prediction for each adverb separately. In each case, we ordered speakers by their average *z*-score on [Adv N] items and by their average *z*-score on [Adv & Adj N] items, and we measured the correlation between these two lists using both Spearman's ρ and Kendall's τ coefficients. The results are presented in figures 6–9.[31]

[29] LaplacesDemon also reports bimodality for one of the four [Adv & Adj N] constructions, *now and future head*, but again with two modes indicating acceptability: 29.7 and 35.7.

[30] Recall that the only adverb that met the prediction of B's analysis that [Adv N] should be less acceptable on average than [Adv & Adj N] was *soon.* However, all tests report unimodality for *soon*, directly contradicting B's analysis.

[31] Acceptance is measured in *z*-scores rather than absolute scores, to counter the effect of different interpretation of scales by different participants. Correlations based on absolute scores are all statistically significantly positive (with $p <$ 0.01 for *once* and $<$ 0.001 for the other three adverbs) and all moderate (*once*, *twice*) or strong (*now*, *soon*).

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

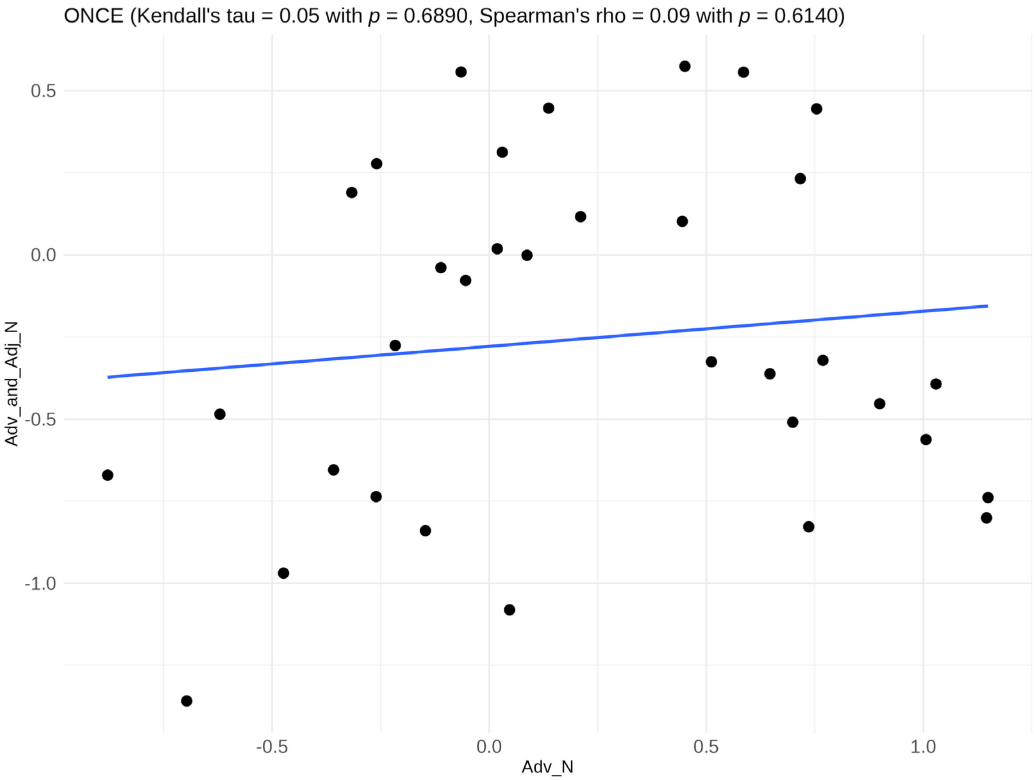


Figure 6
Correlation in respondents' *z*-scores on *once boss* and *once and current boss*; each dot represents a different respondent

All correlations are positive, although in the case of *once* it is negligible and not statistically significant. Other correlations are statistically significant and range from borderline weak/moderate (*twice*),[32] through moderate (*now*), to borderline moderate/strong (*soon*).[33]

4.3 Discussion

Let us take stock. B's (2023) first experiment shows that a considerable number of native speakers accept [Adv N] constructions. B's second experiment was meant to show that they are only accepted as compounds, but it fails to do so: its results are compatible with the usual (under)additive effect of two independent conditions, so it does not demonstrate that *one*-anaphora is impossi-

[32] See https://blogs.sas.com/content/iml/2023/04/05/interpret-spearman-kendall-corr.html on such interpretations of the two correlation coefficients.

[33] Note that the strongest correlation is witnessed for *soon*, which makes it very implausible that *soon appearance* is a compound accepted by a minority of speakers, despite what its diminished acceptability with respect to *soon and certain appearance* might suggest.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

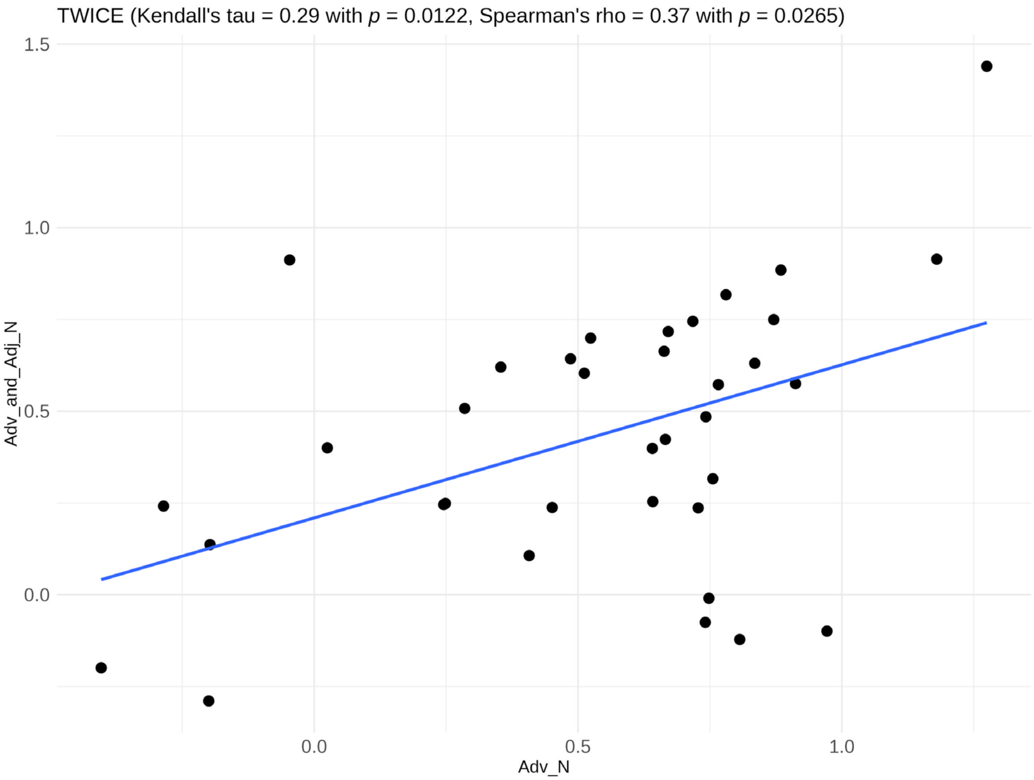


Figure 7
Correlation in respondents' *z*-scores on *twice winner* and *twice or three-time winner*; each dot represents a different respondent

ble in the case of [Adv N].[34] Moreover, our own experiment demonstrates that none of the three predictions of B's analysis is borne out: [Adv N] is not uniformly judged as less acceptable than [Adv & Adj N], judgments on [Adv N] do not have bimodal distributions, and acceptability of [Adv N] correlates with that of [Adv & Adj N]. Hence, experimental evidence unequivocally confirms earlier corpus-based conclusions: [Adv N] constructions with the relevant adverbs are acceptable as syntactic constructions, and not as compounds. There are no "AdvPs in AdjP positions" selectional violations.

The main aim of this article was to refute B's (and B&AK's) claim that constructions such as *the twice or three-time winner* and *the now and future president* involve violations of selectional restrictions, and we believe that this aim has been fully achieved. Nevertheless, we end this section with some speculations on the reasons why prenominal adverbs were found to be less acceptable

[34] Recall the two attested examples of such anaphora in (26)–(27).

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

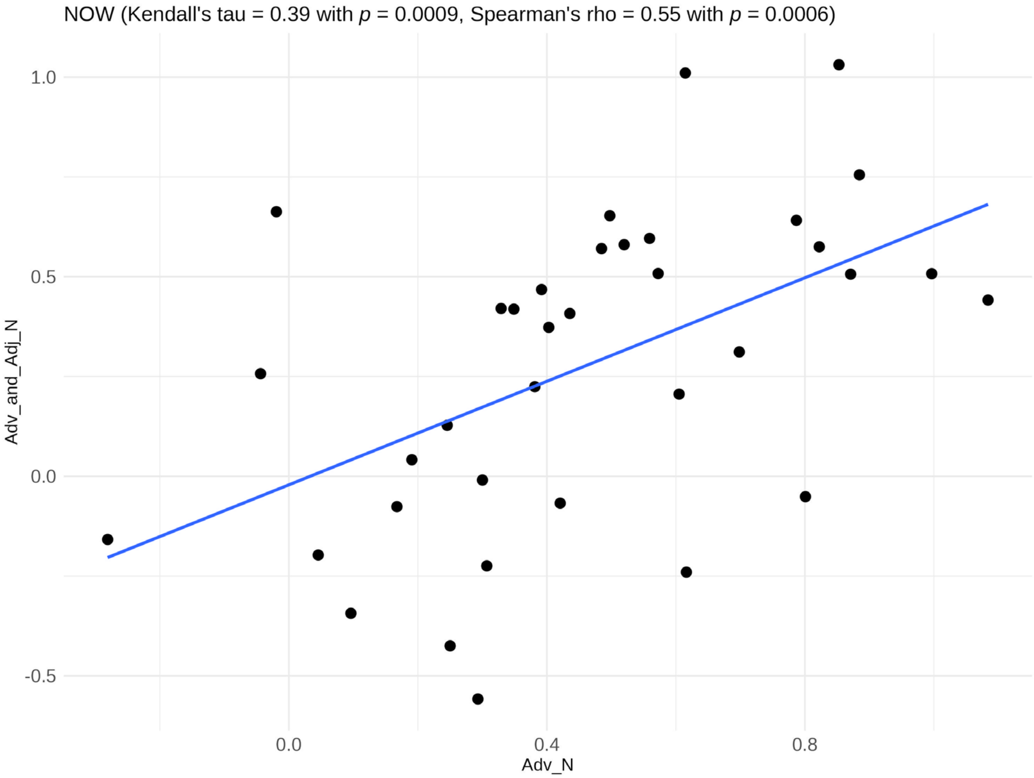


Figure 8
Correlation in respondents' *z*-scores on *now head* and *now and future head*; each dot represents a different respondent

than the corresponding prenominal adjectives, observed in the results of B's experiments (see tables 1 and 2).

The key to explaining this effect seems to be the semantics and pragmatics of these adverbs. They are not exact synonyms of the corresponding adjectives; for example, the prenominal *now* cannot always be replaced with *current*. Consider (11), repeated here as (62), and the attempt to replace *now* with *current* in (63).

(62) In the letter, the *now king* wrote: "Dear Granny, I am sorry that you are ill."

(63) In the letter, the *current king* wrote: "Dear Granny, I am sorry that you are ill."

The context of this sentence is an article in the *Daily Mirror*, written after the death of Elizabeth II, about a letter that Charles III wrote when he was 6 years old. In this context, replacing *now* with *current* is infelicitous, as the two modifiers make different implicit contrasts: *current king* implicitly contrasts with *former king* (i.e., with a different person), while *now king* implicitly contrasts with a different past role of the same person (in this case, with the young Charles). It

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

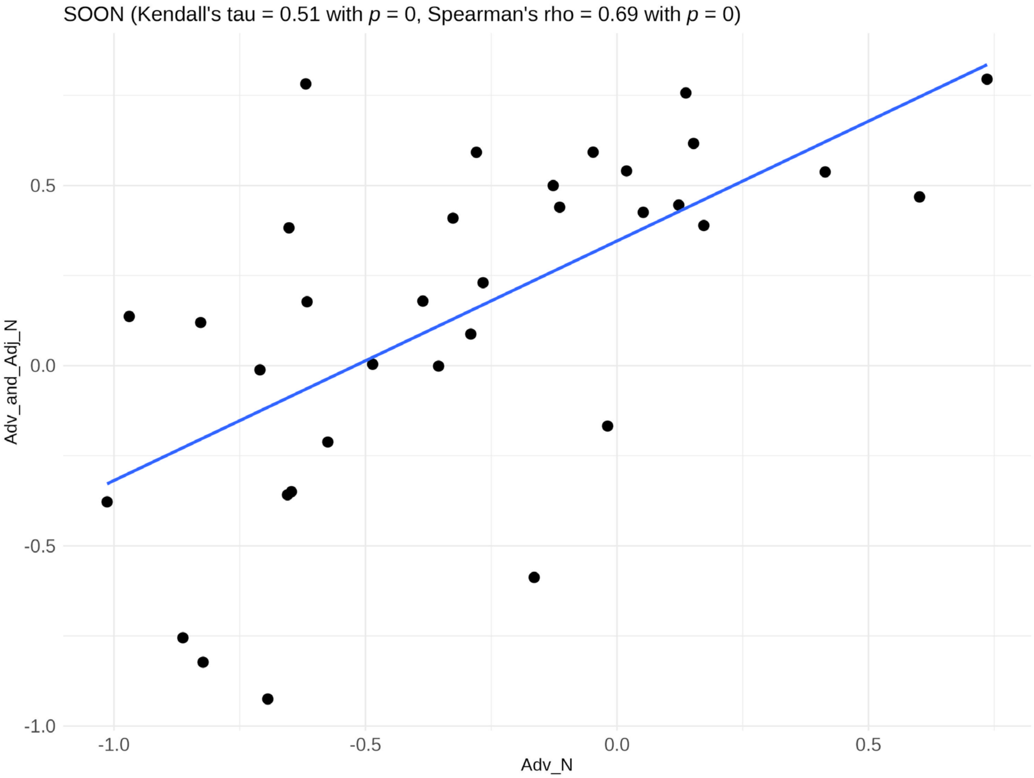


Figure 9
Correlation in respondents' *z*-scores on *soon appearance* and *soon and certain appearance*; each dot represents a different respondent

is this latter contrast that is evoked in (62). Similar differences in implicit contrasts may be detected in the case of *once* (vs. *former*) and *soon* (vs. *future*). We believe that these subtleties in the semantics of the relevant adverbs may be partly responsible for their relatively low frequency in corpora, which in turn negatively impacts their acceptability, given the well-known correlation between frequency and acceptability (see, e.g., Lau, Clark, and Lappin 2017 and references therein).

Second, possibly because of these semantic and pragmatic aspects, prenominal adverbs are more felicitous when they modify a noun whose referent has already been introduced in the discourse. Thus, an out-of-context joke may start with *A former prime minister walks into a bar* . . . , but not with *A once prime minister walks into a bar* . . .[35] In the experiment described in

[35] This intuition is confirmed by corpus proportions of indefinites (which often introduce new discourse referents): 17% for [*a once* N] / [*a*/*the once* N] vs. 41% for [*a former* N] / [*a*/*the former* N]. This difference in proportions is statistically highly significant according to Pearson's chi-square test; $\chi^2(1, N = 2{,}009{,}494) = 849.47$, $p < 0.001$.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

section 4.2, all items with *once* and *twice* have such a referent introduced at the beginning of the sentence (as in (59)), and they are rated significantly higher on average than items with *once* and *soon*, which do not introduce a referent; compare (64) and (65).

(64) They were waiting for the soon appearance of the King.

(65) The once boss of Virgin Records will speak at this event.

We note that all items used in B's experiments also lacked such a facilitating context, which may go some way toward explaining the diminished acceptability of [Adv N].

Finally, it might be the case that different prenominal adverbs have different degrees of acceptability, as a lexical fact best explained from the diachronic and functional perspective.[36] For example *often*, which B claims to be one of the adverbs taking part in selectional violations, is sometimes marked by dictionaries as "archaic" in the prenominal use (e.g., by Dictionary.com), so it may be marginally acceptable to some speakers, the more so, the more the speakers have been exposed to older texts. *Soon* is also clearly less acceptable than other adverbs, according to the results of both B's experiments and ours; as figure 1 shows, its acceptability is at the level of class C standard items.[37] Given the unimodality of judgments, this relatively low acceptability is not a result of the existence of two populations such that [*soon* N] is grammatical for one and ungrammatical for the other (as B would have it); rather, it is a result of genuinely midrange acceptability of some prenominal adverbs. While prenominal adverbs such as *twice* and *now* are significantly more acceptable than *soon*, the end of the scale is probably represented by *then*, which seems to be universally accepted as a prenominal modifier (see section 3.5, especially footnote 18).[38]

Let us finish by noting that, within both [Adv & Adj N] and [Adv N] constructions, there are various collocations.[39] For example, *once and future*—B's main example of [Adv & Adj]—accounts for 95% of matches of the pattern [D *once and* Adj N] in English Web 2020; that is, it is a clear collocation. Within the matches of the more specific pattern [D *once and future* N], the most common noun is *king*, accounting for about 20% of hits. Similarly for some [Adv N] constructions; for example, *soon return* accounts for 45% of matches for [D *soon* N]. As our experiment tested specific lexicalizations of the two constructions, it is possible that some of these lexicalizations were just such collocations. This could explain the large differences between the means of scores on the two constructions in the case of *once* and, especially, *soon*, compared with statistically insignificant differences in the case of *twice* and *now*. For example, it could be that *soon and certain* (or the larger *soon and certain appearance*) is such a collocation, and for

[36] This explanation assumes nonbinary grammaticality. See, for example, Keller 2000, Featherston 2005, 2019, and Lau, Clark, and Lappin 2017 for specific approaches to nonbinary grammaticality, and Francis 2022 for an overview.

[37] As for B's first experiment, we find that [Adv N] items with *soon* have a mean score of 2.48, while with the other three adverbs the means are in the range [3.23, 3.38].

[38] As an anonymous reviewer notes, if the relevant prenominal adverbs are actually adjectives (see sections 3.4–3.5), there might be a coercion process turning relevant adverbs into adjectives, and the applicability of this process may vary across adverbs (and across time and community).

[39] The term *collocation* is understood rather differently in different linguistic traditions (for overviews see, e.g., Evert 2005:chap. 1 and Szudarski 2023:chap. 2). However, a common—on some approaches, defining—feature of collocations is that their corpus frequencies are significantly higher than would be expected from the frequencies of their constituents.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

this reason *soon and certain appearance* is more acceptable than *soon appearance*, with bare *soon*, resulting in the tendency opposite to that observed for *twice*, *now*, and especially *once*, where [Adv N] is more acceptable than [Adv & Adj N].[40] We leave this matter for future research.

5 Conclusion

If *CGEL* is right, coordination is symmetric: X and Y may be conjoined in position P if and only if each of X and Y alone may occur in P. B (2023) disputes this generalization and claims that there are exactly two situations where coordination of X and Y is licensed in P even though X or Y alone is not.

We investigated one of these two purported selectional violations, exemplified by *the now and future president*, and showed that it does not involve any violations and is not an exception to the symmetric nature of coordination. In the process, we demonstrated—using both corpus evidence and experimental methodology—that the relevant [Adv N] constructions are syntactic and generally grammatical, contrary to B's analysis, on which they are at best lexical compounds, acceptable only to a minority of speakers of English. (See also online appendix A for a comprehensive critique of B's analysis.)

In a related publication, Kim and Lu (2024) explain the other purported selectional violation, exemplified by *You can depend on my assistant and that he will be on time*, as the processing effect of grammaticality illusion. If this explanation is confirmed by future research, their results and ours jointly mean that there are no selectional violations in coordination. This in turn means that an important argument for asymmetric theories of coordination collapses, and also that various special mechanisms needed to account for such apparent selectional violations are not needed in syntactic theory. Hence, the effect of refuting the claim that selectional violations are possible in coordination is a more adequate theory of coordination and a leaner grammar.

References

Ameijeiras-Alonso, Jose, Rosa M. Crujeiras, and Alberto Rodríguez-Casal. 2021. multimode: An R package for mode assessment. *Journal of Statistical Software* 97:1–32.
Bard, Ellen Gurman, Dan Robertson, and Antonella Sorace. 1996. Magnitude estimation of linguistic acceptability. *Language* 72:32–68.
Bates, Douglas, Martin Mächler, Ben Bolker, and Steve Walker. 2015. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* 67:1–48.
Bauer, Laurie. 2020. Compounds. In *The Oxford handbook of English grammar*, edited by Bas Aarts, Jill Bowie, and Gergana Popova, 262–280. Oxford University Press.
Bayer, Samuel. 1996. The coordination of unlike categories. *Language* 72:579–616.
Borsley, Robert D. 1994. In defense of coordinate structures. *Linguistic Analysis* 24:218–245.
Borsley, Robert D. 2005. Against ConjP. *Lingua* 115:461–482.

[40] The number of occurrences of the [D *soon and* Adj N] pattern in English Web 2020 is not sufficient to verify this hypothesis, but the results are compatible with it: among the 14 different matches (i.e., after excluding one repeated sentence from the 15 raw matches), only the adjectives *certain* and *literal* occurred twice, and other adjectives each occurred once.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

Brown, J. M. M., Gisbert Fanselow, Rebecca Hall, and Reinhold Kliegl. 2021. Middle ratings rise regardless of grammatical construction: Testing syntactic variability in a repeated exposure paradigm. *PLoS ONE* 16:1–26.

Bruening, Benjamin. 2018a. The lexicalist hypothesis: Both wrong and superfluous. *Language* 94:1–42.

Bruening, Benjamin. 2018b. Word formation is syntactic: Raising in nominalizations. *Glossa: a journal of general linguistics* 3:1–25.

Bruening, Benjamin. 2023. Selectional violations in coordination: A response to Patejuk and Przepiórkowski 2023. *Linguistic Inquiry* Early Access, 1–45. https://doi.org/10.1162/ling_a_00506.

Bruening, Benjamin, and Eman Al Khalaf. 2020. Category mismatches in coordination revisited. *Linguistic Inquiry* 51:1–36. https://doi.org/10.1162/ling_a_00336.

Dalrymple, Mary. 2017. Unlike phrase structure category coordination. In *The very model of a modern linguist*, edited by Victoria Rosén and Koenraad De Smedt, 33–55. University of Bergen Library. https://bells.uib.no/index.php/bells/article/view/1332.

Evert, Stefan. 2005. The statistics of word cooccurrences: Word pairs and collocations. Doctoral dissertation, Institut für Maschinelle Sprachverarbeitung, Universität Stuttgart.

Featherston, Sam. 2005. The Decathlon model of empirical syntax. In *Linguistic evidence: Empirical, theoretical and computational perspectives*, edited by Stephan Kepser and Marga Reis, 187–208. Mouton de Gruyter.

Featherston, Sam. 2008. Thermometer judgements as linguistic evidence. In *Was ist linguistische Evidenz?*, edited by Claudia Maria Riehl and Astrid Rothe, 69–90. Shaker Verlag.

Featherston, Sam. 2009. A scale for measuring well-formedness: Why syntax needs boiling and freezing points. In *The fruits of empirical linguistics.* Vol. 1, *Process*, edited by Sam Featherston and Susanne Winkler, 47–73. Mouton de Gruyter.

Featherston, Sam. 2019. The Decathlon model. In *Current approaches to syntax: A comparative handbook*, edited by András Kertész, Edith Moravcsik, and Csilla Rákosi, 155–186. De Gruyter Mouton.

Featherston, Sam. 2020. Can we build a grammar on the basis of judgments? In *Linguistic intuitions: Evidence and method*, edited by Samuel Schindler, Anna Drożdżowicz, and Karen Brøcker, 165–188. Oxford University Press.

Featherston, Sam. 2021. Response methods in acceptability experiments. In Goodall 2021, 39–61.

Francis, Elaine J. 2022. *Gradient acceptability and linguistic theory.* Oxford University Press.

Gerbrich, Hannah, Vivian Schreier, and Sam Featherston. 2019. Standard items for English judgment studies: Syntax and semantics. In *Experiments in focus: Information structure and semantic processing*, edited by Sam Featherston, Robin Hörnig, Sophie von Wietersheim, and Susanne Winkler, 305–327. De Gruyter.

Gibson, Edward, Steve Piantadosi, and Kristina Fedorenko. 2011. Using Mechanical Turk to obtain and analyze English acceptability judgments. *Language and Linguistics Compass* 5:509–524.

Gibson, Edward, and James Thomas. 1999. Memory limitations and structural forgetting: The perception of complex ungrammatical sentences as grammatical. *Language and Cognitive Processes* 14:225–248.

Giegerich, Heinz. 2009. Compounding and lexicalism. In Lieber and Štekauer 2009b, 178–200.

Goodall, Grant, ed. 2021. *The Cambridge handbook of experimental syntax.* Cambridge University Press.

Hofmeister, Philip, Laura Staum Casasanto, and Ivan A. Sag. 2014. Processing effects in linguistic judgment data: (Super-)additivity and reading span scores. *Language and Cognition* 6:111–145.

Huddleston, Rodney, and Geoffrey K. Pullum. 2002. *The Cambridge grammar of the English language.* Cambridge University Press.

Jakubíček, Miloš, Adam Kilgarriff, Vojtěch Kovář, Pavel Rychlý, and Vít Suchomel. 2013. The TenTen corpus family. In *Corpus Linguistics 2013: Abstract book*, edited by Andrew Hardie and Robbie Love, 125–127. https://ucrel.lancs.ac.uk/cl2013/doc/CL2013-ABSTRACT-BOOK.pdf.

Johannessen, Janne Bondi. 1998. *Coordination.* Oxford University Press.

Keller, Frank. 2000. Gradience in grammar: Experimental and computational aspects of degrees of grammaticality. Doctoral dissertation, University of Edinburgh.

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

Kilgarriff, Adam, Vít Baisa, Jan Bušta, Miloš Jakubíček, Vojtěch Kovář, Jan Michelfeit, Pavel Rychlý, and Vít Suchomel. 2014. The Sketch Engine: Ten years on. *Lexicography* 1:7–36.

Kim, Nayoun, and Jiayi Lu. 2024. Coordination of unlike categories creates grammaticality illusion. *Korean Journal of English Language and Linguistics* 24:52–61.

Kuznetsova, Alexandra, Per B. Brockhoff, and Rune H. B. Christensen. 2017. lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software* 82:1–26.

Lau, Jey Han, Alexander Clark, and Shalom Lappin. 2017. Grammaticality, acceptability, and probability: A probabilistic view of linguistic knowledge. *Cognitive Science* 41:1202–1241.

Levine, Robert D. 2011. Linearization and its discontents. In *Proceedings of the HPSG 2011 Conference*, edited by Stefan Müller, 126–146. CSLI Publications. https://proceedings.hpsg.xyz/article/view/247.

Lieber, Rochelle. 2009. IE, Germanic: English. In Lieber and Štekauer 2009b, 357–369.

Lieber, Rochelle, and Pavol Štekauer. 2009a. Introduction: Status and definition of compounding. In Lieber and Štekauer 2009b, 3–18.

Lieber, Rochelle, and Pavol Štekauer, eds. 2009b. *The Oxford handbook of compounding.* Oxford University Press.

Mächler, Martin. 2024. *Package 'diptest'.* https://cran.r-project.org/web/packages/diptest/diptest.pdf.

Molimpakis, Emilia. 2019. Accepting preposition-stranding under sluicing cross-linguistically: A noisy-channel approach. Doctoral dissertation, UCL.

Munn, Alan. 1993. Topics in the syntax and semantics of coordinate structures. Doctoral dissertation, University of Maryland.

Neeleman, Ad, Joy Philip, Misako Tanaka, and Hans van de Koot. 2023. Subordination and binary branching. *Syntax* 26:41–84. https://onlinelibrary.wiley.com/doi/epdf/10.1111/synt.12244.

Patejuk, Agnieszka, and Adam Przepiórkowski. 2023. Category mismatches in coordination vindicated. *Linguistic Inquiry* 54:326–349. https://doi.org/10.1162/ling_a_00438.

Payne, John, Rodney Huddleston, and Geoffrey K. Pullum. 2010. The distribution and category status of adjectives and adverbs. *Word Structure* 3:31–81.

Przepiórkowski, Adam. 2022. Coordination of unlike grammatical cases (and unlike categories). *Language* 98:592–634. https://muse.jhu.edu/article/864636.

R Core Team. 2012. *R: A language and environment for statistical computing.* R Foundation for Statistical Computing, Vienna. http://www.R-project.org.

Sag, Ivan A., Gerald Gazdar, Thomas Wasow, and Steven Weisler. 1985. Coordination and how to distinguish categories. *Natural Language and Linguistic Theory* 3:117–171.

Statisticat, and LLC. 2021. *LaplacesDemon: Complete environment for Bayesian inference.* https://web.archive.org/web/20150206004624/http://www.bayesian-inference.com/software, R package version 16.1.6.

Szudarski, Paweł. 2023. *Collocations, corpora and language learning.* Cambridge University Press.

Williams, Edwin. 1981. Transformationless grammar. *Linguistic Inquiry* 12:645–653.

Zhang, Niina Ning. 2009. *Coordination in syntax.* Cambridge University Press.

Zhang, (Niina) Ning. 2023. *Coordinate structures.* Cambridge University Press.

Sources of Attested Examples

Most of the attested examples come from English Web 2020: (15), (17)–(20), (24), (28)–(30), (39)–(44), (55a–b), and (56a–b). Other examples were found via Google—specifically:

(9) https://www.bbc.com/news/newsbeat-64209376

(10) https://www.theguardian.com/australia-news/live/2022/sep/10/australia-reacts-to-queen-elizabeth-iis-death-albanese-and-dutton-to-lay-wreaths-at-parliament-house

Linguistic Inquiry Early Access Corrected Proof
https://doi.org/10.1162/ling_a_00548

(11) https://www.mirror.co.uk/news/royals/hand-written-letter-king-charles-29401615
(12) https://www.harpersbazaar.com/celebrity/latest/a42185461/prince-harry-meghan-markle-first-meet-royal-family-netflix/
(13) https://nypost.com/2022/09/23/kate-middleton-prince-william-say-they-saw-rainbows-after-queens-death/
(14) https://www.theroyalforums.com/threads/savoy-and-savoy-aosta-restoration-succession-heirs-and-conflicts-1-ending-2022.2411/page-2
(21) https://www.barbarianrugby.co.nz/wp-content/uploads/BABANewsJun2010.pdf
(22) https://www.transmissionfilms.com.au/films/tommys-honour
(23) https://biz.crast.net/citadels-billionaire-investor-ken-griffin-calls-trump-a-thrice-loser/
(26) https://www.njgsbc.org/files/familyfiles/p512.htm
(27) https://bleedingcool.com/comics/who-will-succeed-boris-johnson-once-future-27-prepares-spoilers/
(34) https://open.spotify.com/episode/6IlbK1uuhmt94rKIPvQMVf
(35) https://themedialine.org/top-stories/bennett-sworn-in-as-israeli-prime-minister-netanyahu-heads-to-opposition/
(36) https://forums.beyondblue.org.au/t5/anxiety/anxious-mother-anxious-family-psychotherapy/td-p/318226
(37) https://www.reddit.com/r/AskReddit/comments/1drlvl/comment/c9t68fk/
(38) https://www.nydailynews.com/entertainment/gossip/bill-hudson-children-oliver-kate-dead-article-1.2274050

Adam Przepiórkowski
Institute of Computer Science, Polish Academy of Sciences
Faculty of Philosophy, University of Warsaw
Department of Linguistics and Philosophy, MIT
przepiorkowski.adam@gmail.com

Agnieszka Patejuk
Institute of Computer Science, Polish Academy of Sciences
agnieszka.patejuk@gmail.com

Appendices to

Prenominal Adverbs, or Apparent Selectional Violations in Coordination

Adam Przepiórkowski and Agnieszka Patejuk

Linguistic Inquiry

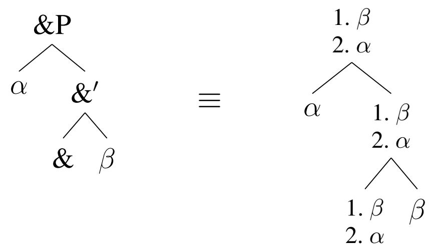
https://doi.org/10.1162/ling_a_00548

Appendix A: Analytical Problems in Bruening 2023

In the case of the apparent "AdvPs in AdjP positions" violations in coordination, B does not retain B&AK's account – criticized at length by P&P – but proposes a completely new analysis. On that analysis, categorial selectional features (C-selectional features, or C-features in short) may be specified using the set notation, as in "[C $\in$ {NP, AdjP}]", i.e., a C-feature that may be checked by an NP (noun phrase) or an AdjP (adjectival phrase), or as in "[C $\notin$ {N, CMPR}]", i.e., a C-feature that may be checked by any category apart from N (noun) or CMPR (comparative).[41] B assumes that modifiers categorially specify the modified constituents. For example, adjectives bear the C-feature specified as "[C $\in$ {N}]", while adverbs such as *once*, *twice*, or *soon* bear the feature "[C $\notin$ {N, CMPR}]". This last specification corresponds to the claim that such adverbs may modify almost any category, with the exception of nouns (recall *the once king* – ungrammatical according to B) or comparatives (see (66a) below from Bruening 2023: 39, apparently originally from Gobeski 2011).[42]

(66) a. She is {two times / *twice} taller than him.
     b. She is twice as tall as him.

Another difference with respect to Bruening and Al Khalaf 2020 is that in Bruening 2023 coordinations are assumed to be headed by the initially feature-less conjunctions,[43] which derive all their features from conjuncts (via Agree); such features are collected on a stack and shared along the spine of the coordinate structure:

(67)


As in Bruening and Al Khalaf 2020, structures are assumed to be built from left to right, and features taken over by the conjunction from the conjuncts are put on a push-down stack as the derivation progresses. This is illustrated schematically in (67), where the first conjunct shares its α features with the conjunction, forming the bottom of the stack, and the second conjunct subsequently adds its β features to the top of the stack.

Given the left-to-right nature of the derivation, there is a stage where the coordination frag-

[41] We assume that, given a finite set of n categories $C_1, \ldots, C_i, C_{i+1}, \ldots, C_n$, the following two specifications are equivalent: "[C $\in$ {$C_1, \ldots, C_i$}]" and "[C $\notin$ {$C_{i+1}, \ldots, C_n$}]". Bruening 2023 is not entirely clear on this matter.

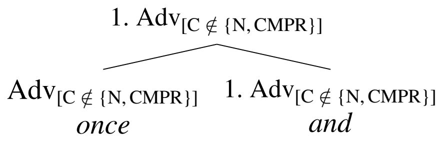
[42] This last claim is most probably false; English Web 2020 contains over 800 examples of *twice* or *thrice*, followed by a comparative form ending in *-er*, followed by *than*, e.g.:

(i) If she's right, she's *twice* stronger than me…
(ii) Horn is *twice* denser than wood so that's a bit less good.
(iii) However, it seemed *twice* smaller than it was before.
(iv) Vegetarians did *twice* better than non-vegetarians (meat-eaters).
(v) Some Home Selling Guides argue that solar powered homes sell *twice* faster than homes using conventional electricity.

[43] The idea that conjunctions are completely featureless seems at odds with the fact that there are conjunctions which have specific C-selectional restrictions; see, e.g., Zhang 2009: §3.3.2 on Chinese. (B cites Zhang 2009 in a different context but does not comment on this incompatibility.)

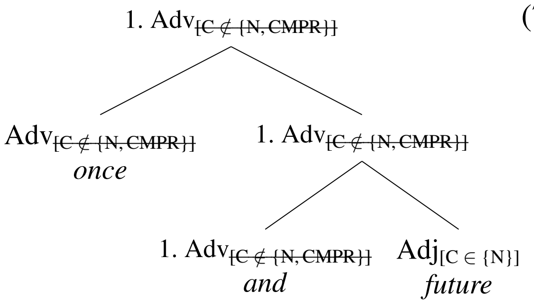
ment *once and* has the following structure:

(68)

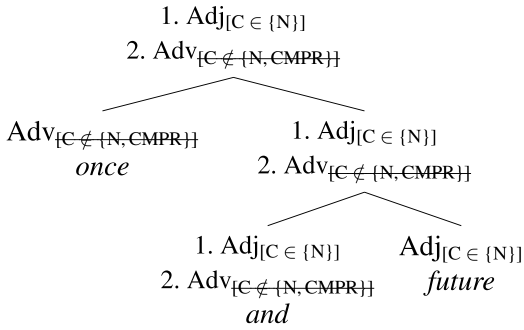


Immediately after *future* is integrated as the second conjunct, the structure of *once and future* is as in (69) (cf. Bruening 2023: 40, ex. (102)).

(69)



(70)



At this stage, the C-feature of the conjunction, $[C \notin \{N, CMPR\}]$, is checked when it is merged with the adjective *future*; this is possible because indeed $Adj \notin \{N, CMPR\}$.[44,45] Subsequently, the categorial and selectional features of the adjective are added, via Agree, to the stack of the conjunction and its projections, resulting in the structure in (70). The only selectional feature left to be checked here is $[C \in \{N\}]$, so *once and future* may be used to modify a noun, unlike *once* alone, which is explicitly precluded from noun modification by the specification $[C \notin \{N, CMPR\}]$.

In the remainder of this appendix, we show that this analysis overgenerates in a number of different ways and, hence, cannot be maintained.

Problem 1: Irrelevance of Order

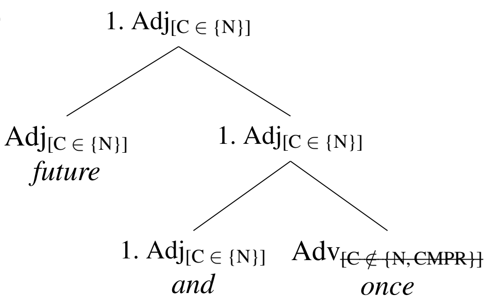
Bruening (2023: 41) claims that, in the case of the reversed order, i.e., *future and once*, the C-feature of *once* "would not be checked off within the coordinate phrase and would percolate to &P", but gives no reasoning to support this claim. We show that this claim does not stand up to scrutiny and that the analysis sketched above does generate structures such as *future and once king*, which B considers ungrammatical.

Consider (71) below – the stage of derivation of *future and once* analogous to the stage of *once and future* in (69).

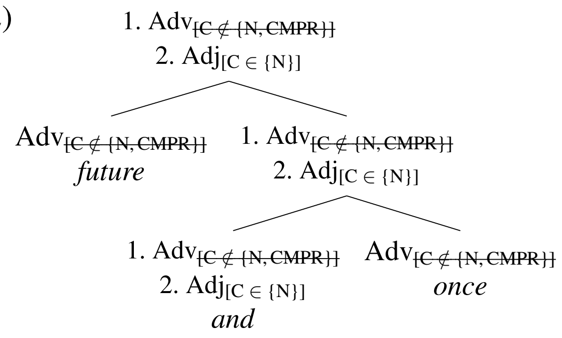
[44]Note that this feature gets marked as checked everywhere it occurs, so such feature specifications cannot be understood as copies, but rather as links to a single "feature specification" object, analogously to the multi-dominance understanding of nodes "copied" in Internal Merge.

[45]Unless the strict ordering "check C-features first, Agree later" is assumed, it seems that nothing in B's setup prevents checking off these C-features already at the stage of *once and*, illustrated in (68). In this configuration, two siblings in the tree select for each other, so these selectional restrictions can be satisfied via mutual selection. Alternatively, one constituent may check off its C-feature against the other, which still results in the checking off of all copies of this C-feature (as noted in the previous footnote).

(71)



(72)



At this stage, the Adv *once* with the specification $[C \notin \{N, CMPR\}]$ merges with the conjunction bearing the category Adj, which checks off this C-feature. It does not matter that the direction of checking in (71) is different than in (69): *once* may both pre- and post-modify (e.g., *once bitten* and *bitten once*). On the assumption that all features of *once* are copied onto the stack, including the checked C-features, the configuration in (72) ensues. However, whether these checked C-features are copied or not, the only unchecked C-feature on the top node is $[C \in \{N\}]$, so *future and once* may modify nouns and, hence, *future and once king* is predicted to be grammatical.

This is the first way in which B's analysis overgenerates with respect to his empirical generalizations.

Problem 2: Recursive Modification

The second way in which B's analysis overgenerates concerns recursive modification, as in (73)–(74).

(73) a terribly white face

(74) a slowly disappearing train

In (73), the adverb *terribly* modifies the adjective *white* and the whole AdjP *terribly white* modifies the noun *face*. Similarly, in (74), *slowly* modifies *disappearing*, and *slowly disappearing* modifies *train*.

The problem with the analysis of Bruening 2023 is that, for any such recursive modification "[[Mod_1 Mod_2] H]", the analysis predicts the grammaticality of "[[Mod_1 & Mod_2] H]". For example, it predicts the grammaticality of (75)–(76).

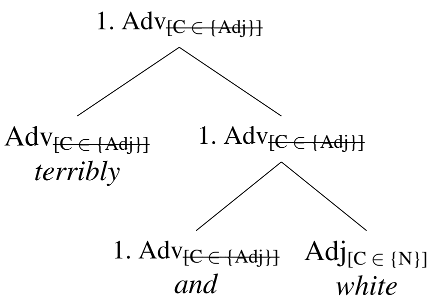
(75) *a terribly and white face

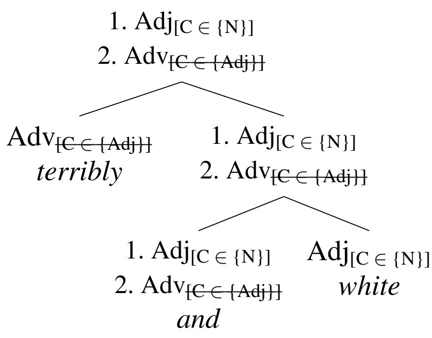
(76) *a slowly and disappearing train

The reason for this is that such coordinations have exactly the same – in all relevant respects – structures as coordinations of the *once and future* type. For example, the structures of the coordination in (75) corresponding to the stages (69)–(70) of *once and future* are shown in (77)–(78).[46]

---

[46]Whatever the selectional restrictions of *terribly*, they must be satisfied by *white*, and whatever the selectional restrictions of *white*, they must be satisfied by *face*. Hence, without any loss of generality, we may simplify C-selectional restrictions of such items and assume that *terribly* is specified as $\text{Adv}_{[C \in \{\text{Adj}\}]}$ and *white* as $\text{Adj}_{[C \in \{N\}]}$.

(77)


(78)


As in the case of *once and future*, the C-features of the adverb are checked off within the coordination, so only the C-features of the adjective are active at the top node, making *terribly and white* a nominal modifier, contrary to the fact that (75) is ungrammatical. The derivation of (76) and other similar ungrammatical examples is analogous.

This is the second respect in which the analysis of Bruening 2023 overgenerates – this time, not just with respect to B's doubtful empirical claims, but more generally.

Other Analytical Loose Ends

In order to account for "AdvPs in AdjP positions", B introduces new mechanisms, without motivating them independently or investigating their consequences. As noted above, the conjunction is supposed to come from the lexicon featureless (and, in particular, categoryless) and to get all its features via Agree with its complement and its specifier. The only independent motivation for this idea is given as "(cf. Murphy and Puškar 2018)" (Bruening 2023: 27). However, mechanisms assumed by B differ from those of Murphy and Puškar 2018 in a number of crucial respects.

First, while Murphy and Puškar (2018) assume that, in nominal coordination, the conjunction has nominal selectional features checked by its complement and specifier, B assumes that conjunctions are featureless. But if so, it is not clear what motivates the Merge of a conjunction and a conjunct: no checking of selectional features takes place at this Merge, and such checking is usually assumed to motivate Merge.[47] This assumption is also explicitly made in Murphy and Puškar 2018: 1221.

Second, in Murphy and Puškar 2018, as elsewhere, what Agree copies from goals to probes are values of particular features such as gender or number, while in Bruening 2023 Agree is supposed to copy complete feature bundles, including the category and features already checked off. The consequences of this radical change in the operation of Agree for the rest of the grammar remain unexplored.

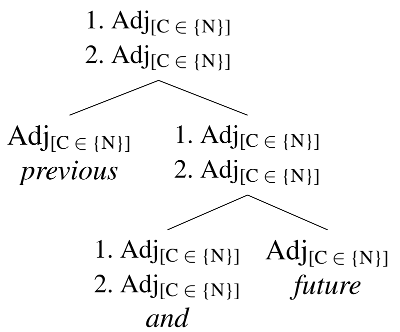
Finally, it is not clear how stacked categories and features interact with the rest of the grammar. B assumes that, when a head such as *become*, with the C-feature "[C ∈ {NP, AdjP}]", combines with the coordination of an NP and an AdjP, such as *a political radical and very antisocial*, then both elements of the stack must satisfy this C-feature, as in (79) from Bruening 2023: 29, (75a).

(79) Danny became$_{[\mathrm{C} \in \{\mathrm{NP, AdjP}\}]}$ [$_{[1:\ \mathrm{AdjP},\ 2:\ \mathrm{NP}]}$ a political radical and very antisocial].

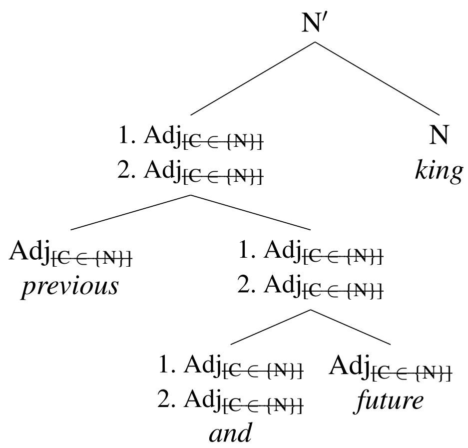
[47]See Zyman 2024 (the "supporting information" section) for arguments against the so-called "Free Merge", not motivated by feature checking.

Hence, one may expect that, dually, C-features of all stack elements should be checked in unison against an item merged with the coordinate structure. For example, in the case of a run-of-the-mill coordination of two adjectives, the stack of the coordinate structure will contain two elements "$\text{Adj}_{[\text{C} \in \{\text{N}\}]}$" as shown in (80), and they both need to be checked by the noun, as in (81), as otherwise one of the C-features would remain unchecked and the derivation would crash.

(80)



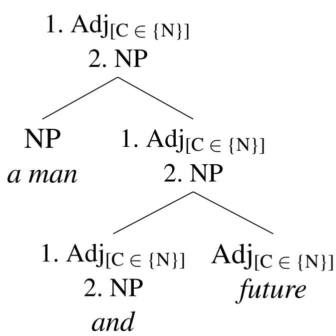
(81)



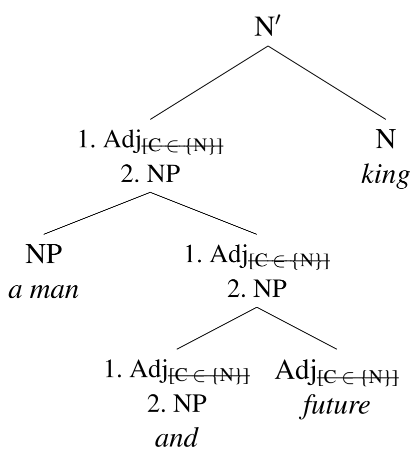
If so, it is not clear how the coordination *once and future*, whose stack contains two elements, only one of which has an active C-feature (see (69)), may combine with the following noun. Here, the feature checking does not work in unison for all stack elements, so the derivation should crash.

Moreover, if it were possible to only look at the top of the stack when checking features of coordination, then it is not clear what would stop coordinating an element without C-selectional features – for example, the NP *a man* – with a modifier, as in the tree in (82); compare this tree with the analogous structure of *once and future* in (70). Once the tree in (82) is merged with a noun, the C-features at the top of the stack are checked, resulting in a well-formed tree, as in (83).

(82)



(83)



It seems that any item without C-features may be the first element in such coordinations.[48]

[48]In particular, since conjunctions do not have any features in the lexicon, it is not clear what would stop them

Hence, this is the third way in which the analysis of Bruening 2023 – rather massively – overgenerates.

Appendix B: Our Experiment

Standard items

Below are all 15 standard items in groups A–E, used in assessing absolute acceptability of stimuli as recommended in Gerbrich et al. 2019.[49]

A.

(84) There's a statue in the middle of the square.
(85) The patient fooled the dentist by pretending to be in pain.
(86) The winter is very harsh in the North.

B.

(87) Before every lesson the teacher must prepare their materials.
(88) Jack doesn't boast about his being elected chairman.
(89) John cleaned his motorbike with which cleaning cloth?

C.

(90) Anna loves, but Linda hates, eating popcorn at the cinema.
(91) Most people like very much a cup of tea in the morning.
(92) The striker fouled deliberately the goalkeeper.

D.

(93) Who did he whisper that had unfairly condemned the prisoner?
(94) The old fisherman took her pipe out of mouth and began story.
(95) Which professor did you claim that the student really admires him?

E.

(96) Historians wondering what cause is disappear civilization.
(97) Old man he work garden grow many flowers and vegetable.
(98) Student must read much book for they become clever.

Exclusion Criteria

The following exclusion criteria were implemented to ensure high quality of responses:

1. Participants must reside in an English-speaking country and report one of the following nationalities: American, British, Irish, Canadian, Australian, or New Zealander (5 participants failed this criterion).[50]

---

from occurring as conjuncts in such configurations, in which case sentences such as *I read an and and wise book* (cf. *I read an interesting and wise book*) would be predicted to be grammatical.

[49] Specifically, we used the versions of these examples in Featherston 2020: 187–188, which sometimes differ a little from those in Gerbrich et al. 2019: 315.

[50] We also asked participants whether they knew other languages (at least) as well as English (10 of the participants included in analysis answered 'yes') and on the number of years spent outside an English-speaking country (M = 0.17, Mdn = 0, Max = 4), but we did not exclude any participants on the basis of this information.

2. Participants must answer at least 9 out of 12 comprehension questions correctly (5 participants).[51]

3. Median response times must be within specific thresholds; see Juzek 2015: 249–251 (cf. Häussler and Juzek 2016: §3.2) for details (3 participants).

4. The average score on severely ungrammatical fillers must be significantly lower than the average score on fully grammatical fillers (2 participants).

5. The average score on severely ungrammatical fillers must be lower than 25 (5 participants).

6. The average score on fully grammatical fillers must be higher than 25 (0 participants).

7. Participants must not be current or former students of linguistics (9 participants).

References

Bruening, Benjamin. 2023. Selectional violations in coordination (a response to Patejuk and Przepiórkowski 2023). URL `https://direct.mit.edu/ling/article-abstract/doi/10.1162/ling_a_00506/115205/Selectional-Violations-in-Coordination-A-Response`, to appear in *Linguistic Inquiry*.

Bruening, Benjamin, and Eman Al Khalaf. 2020. Category mismatches in coordination revisited. *Linguistic Inquiry* 51:1–36. URL `https://direct.mit.edu/ling/article-abstract/51/1/1/96985/Category-Mismatches-in-Coordination-Revisited`.

Featherston, Sam. 2020. Can we build a grammar on the basis of judgments? In *Linguistic intuitions: Evidence and method*, ed. by Samuel Schindler, Anna Drożdżowicz, and Karen Brøcker, 165–188. Oxford: Oxford University Press.

Gerbrich, Hannah, Vivian Schreier, and Sam Featherston. 2019. Standard items for English judgment studies: Syntax and semantics. In *Experiments in focus: Information structure and semantic processing*, ed. by Sam Featherston, Robin Hörnig, Sophie von Wietersheim, and Susanne Winkler, 305–327. Berlin: Mouton de Gruyter.

Gobeski, Adam Michael. 2011. Twice versus two times in phrases of comparison. Master's thesis, Michigan State University.

Häussler, Jana, and Tom S. Juzek. 2016. Detecting and discouraging non-cooperative behavior in online experiments using an acceptability judgment task. In *A blend of MaLT: Selected contributions from the Methods and Linguistic Theories symposium 2015*, ed. by Hanna Christ, Daniel Klenovšak, Lukas Sönning, and Valentin Werner, 73–99. University of Bamberg Press.

Juzek, Tom S. 2015. Acceptability judgement tasks and grammatical theory. Ph.D. dissertation, University of Oxford.

Murphy, Andrew, and Zorica Puškar. 2018. Closest conjunct agreement is an illusion. *Natural Language and Linguistic Theory* 36:1207–1261.

Zhang, Niina Ning. 2009. *Coordination in syntax*. Cambridge: Cambridge University Press.

Zyman, Erik. 2024. On the definition of Merge. URL `https://onlinelibrary.wiley.com/doi/10.1111/synt.12287`, to appear in *Syntax*.

[51] Such comprehension questions were presented at random after a sentence was rated and disappeared from the screen.